

Review on Modulation Formats and Channel Coding in Free Space Optical Communication: Selection Criteria, Trade-Offs, and Emerging Trends

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Abstract—Free Space Optical (FSO) communication systems provide extremely large modulation bandwidths in the optical spectrum (hundreds of THz), enabling potential data rates from hundreds of Mbps to multiple Tbps, far exceeding congested radio frequency (RF) bands. In practical deployments, however, achievable throughput is limited by atmospheric impairments, including turbulence-induced fading, weather-dependent attenuation, beam divergence, and precise pointing requirements. Motivating the need for adaptive optics, hybrid FSO/RF architectures, advanced modulation techniques, and forward error correction coding to maintain reliable high-throughput communication. Nevertheless, FSO links can still provide substantially higher capacity than RF links, particularly in space, due to their much higher carrier frequencies and wider available spectrum. FSO systems span terrestrial, aerial, and space applications, with mitigation strategies tailored to factors such as system complexity, latency, supported applications, power constraints, and size/weight limitations. This survey presents a comprehensive analysis of modulation and channel coding schemes across these scenarios, providing quantitative comparisons previously unavailable. We review key channel impairments, system limitations from experimental studies, and evaluate trade-offs for modulation formats and coding schemes using metrics including bit error rate, outage probability, power and spectral efficiency, coding gain, and implementation complexity. Moreover, we discuss emerging trends and future directions, including AI-driven adaptive modulation and coding, orbital angular momentum multiplexing, and intelligent distributed multi-node systems. This survey serves as both a comprehensive reference and a practical design guide, offering actionable recommendations that map modulation and coding strategies to specific platforms, from terrestrial and uncrewed aerial vehicle (UAV) networks to high-altitude platforms, satellite, and deep-space systems.

Index Terms—Atmospheric channel, error-correction codes (ECC), FSO, non-terrestrial networks (NTN), optical modulation, terrestrial networks.

I. INTRODUCTION

FSO communication is a system that employs an optical carrier of the electromagnetic spectrum to provide secure, safe, cost-effective, and reliable communication solutions [1]. The concept of FSO communication can be traced back to Alexander Graham Bell's Photophone in 1880, which successfully transmitted voice over a 200 m path using modulated sunlight [2]. Interest in FSO was revived in the 1960 s with the invention of lasers and laser diodes (LDs), leading to landmark demonstrations such as MIT's 30-mile television link using lasers, and North American Aviation's early experiments with TV-over-laser transmission. In 1975, Nippon Electric Company (NEC) deployed the world's first commercial FSO system: a 14 km full-duplex He–Ne laser link [3]. Subsequently, military applications spurred extensive research, and recent advances in optoelectronic components have enabled a wide range of experimental FSO field trials. One of the primary advantages of FSO communication is its access to a large, license-free bandwidth, which enables the potential for extremely high data rates. As the world becomes increasingly interconnected, driven by the rapid expansion of 5G/6G infrastructure, the Internet of Things (IoT), and large-scale data services, FSO is positioned as a key enabler for bandwidth-intensive front-haul and back-haul links in next-generation networks.

In addition, hybrid FSO/RF systems or FSO backup systems for on-demand applications must have low deployment cost and offer scalability with low maintenance, especially in frequent, diverse, or high turbulence levels. Hybrid FSO/RF systems exploit link diversity by combining the high-capacity advantages of optical links with the robustness of RF links to maintain connectivity under adverse atmospheric conditions, with overall performance largely determined by switching strategies, outage prediction, and hysteresis control [4], [5]. Switching decisions are commonly based on SNR-based thresholds, which offer low complexity and are widely adopted in practice, BER-based criteria, which directly align link selection with QoS requirements at the expense of real-time error estimation, and predictive switching, which anticipates FSO degradation to reduce packet loss and unnecessary transitions [6], [7]. Predictive schemes rely on outage prediction algorithms that can be broadly classified into statistical forecasting methods with modest computational cost but limited robustness to rapid weather changes [8], and

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machine-learning-based approaches, which achieve higher prediction accuracy under non-stationary conditions at the cost of increased computational and hardware requirements [9]. To avoid rapid oscillations between links when channel quality fluctuates near decision thresholds, hysteresis modeling is essential, with properly designed hysteresis margins shown to significantly reduce ping-pong effects while preserving high FSO utilization. Therefore, practical implementation remains constrained by switching latency, buffer management during link transitions, hardware synchronization between heterogeneous transceivers, and the spectrum efficiency trade-off introduced by RF backup links.

Nevertheless, FSO systems face several significant foremost among them being their susceptibility to adverse weather conditions, including fog, rain, and atmospheric turbulence, which can cause severe signal degradation or even complete link outages. Moreover, in mobile applications such as satellite and airborne communications, precise alignment between the transmitter and receiver is critical, making pointing, acquisition, and tracking (PAT) systems indispensable [10]. In addition, beam divergence and power limitations impose constraints on the communication range, making it challenging to maintain satisfactory signal quality over extended distances without the deployment of sophisticated mitigation techniques. Despite these limitations, FSO communication remains a highly promising technology, particularly for applications that require rapid deployment, high data rates, and secure transmission over short to medium distances [11].

To address these challenges, many research efforts have focused on system designs that are robust to a wide range of channel conditions. Modulation and coding, in particular, are key enablers in improving link reliability, spectral efficiency, and power efficiency. From advanced modulation schemes like quadrature amplitude modulation (M -QAM) and orthogonal frequency division multiplexing (OFDM) variants to sophisticated channel coding techniques such as Low-Density Parity-Check (LDPC) codes and Turbo codes, researchers have proposed and evaluated numerous solutions. Each approach comes with its own balance of complexity, implementation cost, and performance gains, creating a multi-criteria design trade-off that system designers must navigate. Furthermore, design considerations can vary significantly depending on the operating environment, whether the system is deployed over short terrestrial links, installed on airborne platforms, or integrated into satellite communication payloads.

Despite the recognized importance of modulation schemes and channel coding techniques in FSO communication, a comprehensive survey that systematically examines how these approaches vary across different FSO deployment scenarios is notably absent. Existing surveys typically focus on a single domain or a subset of physical-layer techniques, such as modulation formats only or channel modeling only. As a result, system designers facing a specific scenario lack a unified reference that compares which combinations of modulation, coding, and enhancement techniques are most suitable under given constraints. This survey addresses this gap by adopting a scenario-driven and decision-oriented perspective. Instead of merely cataloging

existing schemes, we systematically compare terrestrial, aerial, and satellite FSO links along key dimensions, including link distance, mobility, pointing errors, channel variability, turbulence severity, visibility constraints, power limitations, beam divergence, regulatory requirements, deployment complexity, and latency. Building on this characterization, we evaluate mainstream modulation formats and channel coding schemes using a multi-criteria framework that jointly considers simplicity/complexity, BER performance, power and spectral efficiency, robustness to burst and random errors, and decoding latency. These evaluations are visualized through radar charts, providing an intuitive tool for selecting candidate schemes tailored to each scenario.

Compared with prior surveys (Table I), this work advances the field through the following critical operational distinctions that address a concrete gap in the design literature:

- 1) **Scenario-aware modulation-coding co-optimization:** While recent surveys and optical communication tutorials catalog modulation formats and coding schemes in isolation, this work quantitatively evaluates modulation-coding combinations under deployment-specific constraints. Key metrics: BER, outage probability, power efficiency, spectral efficiency, coding gain, and complexity are compared using normalized multi-metric radar charts, enabling engineers to directly map link characteristics (distance, mobility, atmospheric profile) to recommended configurations.
- 2) **Comparative cross-domain performance evaluation:** Prior domain-specific surveys (e.g., deep-space communications, terrestrial FSO) excel within their scope but do not enable cross-domain comparison. This work systematically contrasts the dominant channel impairments across terrestrial, aerial, and space FSO links, showing how factors such as geometric loss, pointing jitter, and scintillation dominate in different distance regimes and qualitatively assesses modulation and coding format suitability per domain, overcoming the single-environment scope of prior surveys.
- 3) **Integration of practical constraints and realworld validation:** This work explicitly incorporates deployment realities—including power budgets, beam divergence limits, pointing error margins, regulatory latency bounds, and receiver complexity constraints—as design boundaries, an aspect missing from general surveys. These constraints are grounded in an extensive collection of experimental field trials and operational systems (e.g., LLCD, LCRD, EDRS, Starlink, Project Loon, and terrestrial multiTb/s demonstrations), which serve as validation cases for the proposed design framework.
- 4) **Emerging trends and security-aware future directions:** Incorporating physical-layer security as a deployment-dependent design dimension across terrestrial, aerial, and space FSO systems, linking modulation and coding choices to interception resilience and secrecy margins under practical constraints. Additionally, systematically examining emerging trends, including AI-driven adaptive modulation, polarization, and OAM multiplexing, and intelligent ML-based decoding strategies, evaluating their

TABLE I
COMPARISON OF THIS WORK WITH EXISTING FSO SURVEYS: DISTINGUISHING METHODOLOGICAL DEPTH AND DESIGN APPLICABILITY.

		[14] 2014	[15] 2015	[16] 2021	[17] 2024	[18] 2025	[19] 2025	This Work —
Deployment Scope		Terrestrial	Terrestrial, space	General FSO	General FSO	General FSO	Outdoor FSO	Terrestrial, aerial, space
Technical Coverage	Channel Modeling	✓	✓	✓	✓	×	✓	✓
	Channel Coding	✓	✓	×	✓	×	×	✓
	Outage Probability	✓	×	✓	×	×	✓	✓
Modulations Performance (Analytical Expressions)	Power Efficiency	×	Partial	✓	×	×	×	✓
	Spectral Efficiency	×	Partial	✓	✓	×	×	✓
	BER	×	×	✓	×	×	×	✓
Analysis Methodology	Modulation-Coding Co-Design	×	×	×	×	×	×	✓
	Design Decision Trees	×	×	×	×	×	×	✓
	Constraint-Performance Mapping	×	×	×	×	×	×	✓
Enhancing Techniques	OAM	×	✓	×	✓	×	✓	✓
	Machine Learning	×	×	×	✓	×	✓	✓
	Adaptive Transmission	✓	✓	×	✓	✓	✓	✓
Comparative Analysis		×	×	×	×	×	×	✓
Environment Security Implication		×	×	×	✓	✓	✓	✓

robustness, efficiency, and implementation feasibility to outline a structured roadmap for next-generation secure FSO networks.

Paper Organization

The remainder of this survey is organized as follows. Section II introduces the main categories of FSO communication links, distinguishing between terrestrial and non-terrestrial systems, with dedicated discussions on the constraints, system requirements, and recent case studies for each application scenario. Section III reviews the fundamental principles of laser communication systems, including system operation and channel modeling considerations relevant to different deployment environments. Section IV provides a comprehensive examination of modulation techniques for FSO communication, with the key performance metrics followed by detailed analyses of baseband, passband, and multi-carrier modulation formats. Section V then surveys channel coding techniques, emphasizing the technical principles, advantages, and their selection criteria. Section VI synthesizes insights across the previous sections by discussing the trade-offs that govern modulation and coding choices with further design guidelines to assist in selecting suitable physical-layer schemes under different operational constraints. Section VII highlights the security implications of modulation and coding in FSO systems, including vulnerabilities and mitigation strategies. Section VIII outlines emerging trends and future research directions, covering advances in modulation formats and channel coding such as orbital angular momentum (OAM), machine learning (ML) and adaptive schemes motivated by evolving system requirements and technological capabilities. Finally, Section IX concludes the paper.

II. FSO COMMUNICATION LINKS

This section compares terrestrial, aerial, and space-based FSO communication platforms by examining their distinct operational constraints and design requirements. It also reviews representative real-world deployments in each category, thereby linking theoretical models with practical performance and highlighting the maturity and readiness of FSO technologies for emerging applications.

A. Terrestrial Systems

Terrestrial FSO links typically span from a few hundred meters up to several kilometers, providing high-bandwidth, low-latency connectivity in urban environments [18]. Deployed between buildings, towers, or across campus grounds, these systems complement existing fiber and RF infrastructure by offering rapid “last-mile” and backup connectivity at a fraction of the installation cost [19]. In cellular networks, they facilitate high-capacity backhaul between base stations and core switches, and in disaster recovery, they enable temporary on-demand links when conventional infrastructure is compromised. Moreover, FSO can traverse challenging terrain, such as waterways or urban canyons, where trenching fiber is impractical or prohibitively expensive. Despite these advantages, terrestrial FSO performance is highly sensitive to atmospheric conditions. Fog, rain, haze, and air pollution introduce absorption and scattering losses that reduce signal strength and limit link availability. Consequently, system design must balance aperture sizing, modulation schemes, and adaptive power control to mitigate weather-induced attenuation and maintain reliable operation. Laser safety regulations are another limitation that the terrestrial

FSO system must comply with, which reduces the transmitted optical power, affecting the coverage and range [20].

Authors of [21] offered an experimental study to observe and quantify air turbulence-induced changes in the beam AOA, which accounts for oscillations. In order to achieve high-speed beam tracking and steering, this work has created an optical antenna with a fine positioning mirror (FPM), which reduces optical power fluctuations in the received beam linked to the single-mode fiber (SMF). A hybrid solution that combines FSO with a backup RF system is advised in order to balance the dependability of RF with the high speed of FSO. In a building-to-building configuration, [22] evaluated an ultra-broadband optical wireless system. It was shown that an FSO system with wavelength multiplexing, coherent detection, and differential phase-shift keying (DP-QPSK) format could support 1.6 terabits per second bandwidth over an 80-meter free-space link and 40-kilometer optical fiber propagation. In order to forecast attenuation for FSO systems, [23] collected and analyzed visibility data over a three-year period in Malaysia's tropical environment. It was discovered that dense haze had a greater impact on visibility and FSO system function than heavy rainfall. Seasonal forest fires and a lot of rain worsen visibility problems. Another terrestrial FSO case study was on the security of non-line-of-sight (NLoS) terrestrial FSO communication systems against eavesdroppers [24]. It showed that with high visibility, security hazards are negligible, but under low visibility, they are substantial. Additionally, the study evaluates how the position of the eavesdropper impacts secrecy capacity and calculates the secrecy capacity outage probability for FSO systems influenced by turbulence. The results clarify that the inherent security of FSO links—due to narrow, highly directional beams—is conditional. While LOS links remain secure under clear atmospheric conditions, low visibility increases scattering, allowing eavesdroppers outside the beam to receive sufficient signal energy, thereby reducing secrecy capacity. Using a controlled chamber, the study in [25] investigated the performance of FSO communication systems in dusty settings. The experimental results showed that, in moderate and light dust conditions, FSO communication may be successfully created across short distances. Nevertheless, these links' performance was severely affected by thick dust. Furthermore, at comparable visibility levels, dust was shown to produce almost seven times higher signal attenuation than fog. The attenuation coefficients for an over-the-sea FSO system are estimated empirically in this research [26] by taking humidity, wind speed, and temperature into account. This paper [27] describes a field test of a 1.8 km FSO link that uses optimal forward error correction (FEC), coherent optics, wavelength division multiplexing (WDM) to achieve turbulence mitigation, and over 4 Tbps throughput. An increase in average optical intensity and a 16 % reduction in scintillation index were seen in [28] where they conducted experiments using a 19-aperture optical system for a 2 km horizontal transmission to overcome turbulence. The work in [29] presented an adaptive power pre-compensation system that makes use of an affordable, long-range (LoRa) RF feedback connection to provide 100 Gbps coherent FSO transmission in a 1.8 km field experiment. When the LoS for long-range terrestrial FSO is a problem, implementing a hybrid FSO/RF link, where

the RF link acts as a backup for the FSO, is a probable solution [30]. To conclude, terrestrial FSO systems are now past the research phase and are already deployed for several applications due to their low complexity and deployment concerns. However, maintaining LoS and atmospheric channels having absorption, scattering, and attenuation are the main disturbing challenge that needs to be overcome to achieve reliable visibility and a high-throughput communication link.

In terrestrial FSO, the absence of direct LoS can limit its practical deployment due to obstacles near the ground or no feasible link between the transmitter and receiver [18]. To address this challenge, non-terrestrial networks (NTNs) as aerial and space communication, complement terrestrial platforms and provide coverage where terrestrial deployment is not efficient, expensive, or difficult to implement in hard-to-reach areas.

B. Non-Terrestrial Systems

Equipping flying ships, drones, and satellites with FSO systems will offer a higher probability of LOS and open the way for a huge number of applications transmitting fast and reliable data [31]. Non-terrestrial systems can be categorized based on the altitude and platform of the communication system used, in aerial and space systems. Uncrewed aerial vehicles (UAVs) or high-altitude platforms (HAPs) commercialization has made it more feasible to load flexible air-borne vehicles with communication platforms for several applications. If they have an altitude of 17–22 km above the Earth's surface, they are called high HAPs. At altitudes less than 10 km, they are referred to as low altitude platforms (LAPs) [32]. Space communication, by its nature, has different links depending on the application. Deep space (DS) communication occurs when the communication link is established over very long distances, such as with a station on the moon or planets. When satellites communicate with each other on the same orbit or different orbits, these are referred to as inter-satellite links (ISL). Satellites can also act as relays between HAPs or between a HAP and a ground station. In such a topology, it is called an inter-orbital link (IOL).

1) *Aerial Systems:* Aerial platforms with a variety of altitudes using UAVs and HAPs have emerged as promising solutions to complement the existing terrestrial platforms. These technologies enable OWC systems to extend their coverage area and improve overall performance in terms of capacity, reliability, and energy efficiency [33]. Integrating air-based platforms into FSO or RF systems represents a significant advancement toward achieving ubiquitous connectivity in future wireless networks like 6G. In addition, aerial FSO systems can quickly restore communication in areas where infrastructure has been damaged or when there is a temporary communication need. Different mobile or quasi-stationary aerial platforms have been deployed in FSO, and based on the operational altitude, each has its own constraints, performance, and application. HAPs usually can accommodate more equipment, have longer endurance, and offer a higher field of view (FoV) due to the high altitude, but they are quasi-stationary position. On the other hand, a relatively lower-cost variety of LAP platforms, such as drones and gliders, offer more freedom of movement but with limited power and

weight capacity [34]. Aerial FSO links have various topologies, such as horizontal links (between platforms at the same height), downlinks (between a HAP/LAP and a ground station), and uplinks (from a LAP to a HAP or a satellite).

Whether it's a vertical link where there is human rehabilitation or a horizontal link where other flying vehicles exist, aerial FSO systems must adhere to laws governing laser safety and interference with other communication systems. This involves ensuring that the laser beams are safe for human beings and do not interfere with current radio frequency communication systems [35]. Another regard is platform instability because of movements that can lead to beam misalignment, thus requiring robust stabilization mechanisms. The dynamic position is another challenge that may introduce additional atmospheric variation, which makes modeling the wholesome channel crucial. Also, power constraints are another big consideration as the available power resources especially on small UAVs can restrict the use of high-power FSO systems [36].

Turbulence in the Earth's atmosphere generates fast oscillations in the index of refraction, resulting in beam wandering, scintillation, and loss of signal coherence. These effects can significantly impair the performance of optical communication systems, resulting in increased BER and lower data throughput [37]. To explore the effect of turbulence, misalignment, and cross-talk, multiplexing of OAM was implemented in [38]. They achieved 80 Gbit/s for a 50 m link between a ground station and a UAV. To overcome the SWaP disadvantage of a small aerial platform, [39] used an electro-absorption modulator (EAM) with a modulating retro-reflector (MRR) on the UAV. The setup achieved a rate of 500 Mbps with a range of 560 m for 7.6×10^{-4} BER. Another SWaP-abiding prototype was presented in [40]. It doesn't only support quantum key distribution (QKD) links but also has a mechanism to track the ground station. In the work of [41], [42], analytical formulas for a pointing error for the FSO system were derived based on a UAV hovering channel. Their findings are for different turbulence channel scenarios and evaluated BER and outage probability. Results showed that HD detection outperforms DD detection in mitigating turbulence effects. A demonstration of a lightweight UAV that supports bidirectional communication is done in [43]. Using coherent FSO communication for a range of 10-20 km, a 100 Gbps rate link was observed while utilizing a large beamwidth to mitigate vibrations in the communication link. A 10 Gbps, full-duplex, 30 km range project in [44] used a tethered UAV (TUAV) that offers the ability to carry high-rate laser communication terminals. For higher altitudes, project Loon [45] used HAP balloons to study channel effects on FSO to provide LTE internet connectivity as an augmented solution to terrestrial networks.

2) *Space Systems*: Satellites can exist in various orbits around the earth and depending on their height they are categorized into very low-Earth orbit (VLEO) satellites having an altitude range of 100-450 km, low-Earth orbit (LEO) satellites with up to 2000 km altitude, medium-Earth orbit (MEO) satellite with 2,000-20,000 km altitude and geostationary earth orbit (GEO) satellite with 35,786 km altitude [46]. While the maximum propagation distances between different satellite pairs are as follows: 5,860 km for LEO-LEO, 18,460 km for

MEO-MEO, and 45,000 km for LEO-GEO satellites [47]. In higher orbital altitudes, the communication system exhibits significant resilience against atmospheric attenuation and turbulence. This is because the majority of the signal path is through the vacuum of space, where atmospheric effects are minimal. Atmospheric attenuation is primarily a concern for the lower portions of the signal path near the Earth's surface [48].

PAT is a major concern in space FSO to ensure continuous alignment between satellites to maintain the optical link over long distances. In addition, the Doppler shift induced by the satellite motion needs compensation to maintain signal coherence [49]. Background radiation is another obstacle due to the space environments introducing background radiation that can interfere with optical signals [50]. Due to the massive range between the space transmitter and receiver, latency is introduced, thus affecting the choice of served applications, modulation, and ECC.

NASA's Lunar Laser Communication Demonstration (LLCD) [51] was reported to be the first communication between a ground station and a lunar orbit satellite for both uplink and downlink. Using PAT, time division multiplexing, and turbo codes, they achieved high rates while minimizing BER in a dusty channel environment. With the goal of determining how atmospheric conditions affect laser communication performance, NASA's Laser Communications Relay Demonstration (LCRD) mission started its Experiment Program in 2022 [52]. The geosynchronous payload of the LCRD consists of two linked laser terminals that transport data between optical ground stations in Hawaii and California. Another use case was to enable quick data transfer for Earth observation missions and other space-related operations [53]. The European Data Relay System (EDRS) employs optical communication to deliver high-speed data relay services between GEO satellites and ground stations. Moreover, SpaceX's Starlink project aims at lowering latency and improving worldwide internet coverage by integrating optical communication for high-speed data lines between satellites in orbit [54]. Using satellites to overcome the limitations of ground-based QKD systems that provide secure communication was presented in [55]. In which a LEO satellite achieved a rate of about 20 times higher than optical fibers over a distance of up to 1,200 kilometers. The CubeSat Laser Infrared Crosslink (CLICK) project demonstrated a 20 Mbps SWaP LEO cubesat for communication [56], [57]. Future MEO satellite constellation; O3b mPOWER is being developed by SES and is aimed to provide service for cloud networks, cruise ships, offshore energy, enterprises, mobile backhaul, maritime, aviation, and cloud services [58]. Marking a significant milestone in deep space optical communications, the deep space optical communications (DSOC) led by NASA's Jet Propulsion Laboratory established a two-way communication between Earth and up to Mars distance [59]. These initiatives, each addressing particular difficulties, show the promise of laser communications in space FSO across several orbital regimes.

To ensure the reliable transmission of payload data, tracking data, and control signals, satellite communication

TABLE II
COMPARISON AND SUMMARY OF FSO SYSTEMS CHALLENGES

Aspect	Terrestrial	Aerial	Satellite
Link Distance	Shorter, constrained by obstacles and terrain	Variable, ranging from LAPs to HAPs	Extremely long (thousands of km between satellites or deep space)
Mobility	Low (mainly stationary)	High (moving platforms like UAVs, aircraft)	Extremely high (orbital high velocities)
Pointing Errors	Minimal due to vibration (fixed transceivers)	Significant due to movement	Critical; requires precise PAT
Channel Variation	Stable	Dynamic and variable	Highly variable; different for ISL, uplink and downlink
Turbulence	Weak to strong near ground depending on the environment	Significant at higher altitudes	Can be negligible compared to attenuation; occurs mainly during atmospheric entry/exit process
LoS visibility and	Challenging due to buildings, terrain	Easier LoS, few physical obstacles	Clear LoS required between satellite and ground/satellites
Power Constraints	Stable power availability	Limited by onboard power resources	Strict power limits
Beam Divergence	Less critical (shorter distances)	A concern due to platform motion and distance	Critical due to long distances
Regulatory Concerns	Fewer regulatory concerns	Strict aviation regulations, safety requirements	Strict international regulations and coordination between space agencies
Deployment Complexity	Low, fixed infrastructure-based deployment	High due to mobility, SWaP, and PAT issues	Very high; due to launch logistics, orbital maintenance, and precise PAT.
Latency	Low	Moderate	High latency due to large distances (especially for GEO satellites and DS)

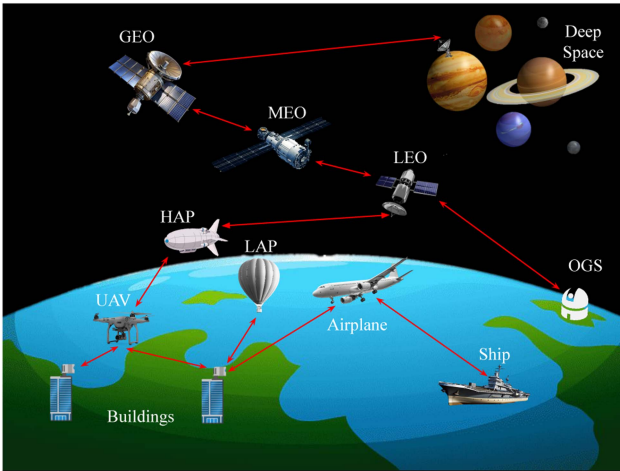


Fig. 1. Different Environments for Optical Communications.

systems require receivers that are both highly sensitive and energy-efficient. These characteristics are crucial for maintaining signal integrity over vast distances and through the relatively weak signals that can be received from satellites. Space optical communications necessitate robust modulation schemes and FEC techniques [60]. The primary reason for this is the significant reduction in received optical power intensity, which occurs due to the vast distances involved and the resulting scarcity of signal photons. Robust modulation schemes help to maximize the efficiency of data transmission, while FEC techniques are essential for correcting errors that arise from weak signals and any potential interference.

Fig. 1 illustrates various platforms used in FSO wireless outdoor communication systems, encompassing both terrestrial and non-terrestrial scenarios, while Table II summarizes the key challenges associated with each platform and highlights their distinguishing characteristics.

III. FUNDAMENTALS OF LASER COMMUNICATION SYSTEMS

Developments in laser diodes are what have enabled transmitters to alter their output power intensity in response to variations in the input signal [61]. This variation can be used to encode information on an optical carrier wave; such process is referred to as modulation. FSO systems can employ intensity modulation/direct detection (IM/DD) or coherent modulation/coherent detection (CM/CD) [62]. IM/DD represents information using the intensity (amplitude) of an optical carrier signal [63]. On the transmitter side, the optical signal is generated using an LD or a light-emitting diode (LED). A photodetector (PD), such as a Positive-Intrinsic-Negative (PIN) diode or an Avalanche Photodiode (APD), is used in the receiver to transform the optical signal received into an electrical signal. The electrical signal is then amplified, filtered, and processed using various signal processing techniques [64]. Because of their low cost and ease of installation, IM/DD has become commonly employed in both FSO and fiber optic systems. Unlike IM/DD, coherent communication encodes information utilizing both the amplitude and phase of the optical carrier. A typical coherent digital optical receiver has two subsystems. The receiver front-end is made up of an optical front-end, a transimpedance amplifier (TIA), and an analog-to-digital converter (ADC), which is followed by a digital signal processor (DSP). To generate the reference signal

at the optical front end, a monochromatic laser serves as the local oscillator (LO). The optical and LO signals are combined and supplied to the photodetector. The photodetectors produce I and Q components of the message signals. TIAs convert the output currents to voltage, which is then digitized by a series of ADCs. The produced signals are then sent to the second subsystem, which consists of a sequence of DSP algorithms. There are two ways to build coherent receivers [31]. If the LO frequency matches the received signal, it is referred to as homodyne coherent detection. While in heterodyne coherent detection, the LO frequency differs from the received signal frequency. Then the message signal must be recovered using a second RF down-conversion. Quantitative comparisons of homodyne and heterodyne detection reveal distinct trade-offs across FSO platforms. Homodyne detection theoretically offers a 3 dB sensitivity advantage under ideal phase matching [65], but this gain diminishes under practical turbulence-induced phase noise, often reducing to 1-3 dB in terrestrial links even with adaptive optics [66]. Heterodyne detection, while exhibiting a 1-2 dB penalty relative to ideal homodyne, demonstrates superior robustness to phase fluctuations and vibration-induced misalignment, making it the preferred choice for aerial and satellite platforms [67]. For instance, satellite FSO heterodyne systems can achieve performance approaching the quantum limit when properly phase-locked, whereas homodyne implementations require complex phase-tracking loops that significantly increase terminal complexity and power consumption [68]. Advanced modulation methods such as quadrature amplitude modulation (QAM) and phase shift keying (PSK) can be used with CM/CD. Other advantages of CM/CD include increased receiver sensitivity and the simplicity of applying digital signal processing for post-processing. Both IM/DD and CM/CD have been considered in the literature on FSO systems [11], [69].

Among the many advantages of laser-based FSO communication is its large modulation bandwidth, which permits high data rates because of the broad frequency range of the optical carriers. FSO is advantageous in terms of avoiding the congestion problems common in RF communications by operating in an unlicensed spectrum. Because it requires less setup time and can be easily deployed and redeployed, it is also a cost-effective deployment option, offering equivalent data rates at a lower cost than RF systems. Moreover, the narrow optical beam of an FSO link provides strong spatial isolation and inherent security, significantly reducing interference [70]. However, the operation of FSO is highly dependent on atmospheric conditions, which could degrade communication, especially in severe weather [71].

A. Basic Principles for FSO Communication System

To transmit data through the atmospheric channel, the transmitter first performs necessary pre-processing, such as modulation and forward error correction coding, and uses a driver circuit to bias the laser above its threshold. FSO links commonly operate in the 850 nm, 1250 nm, or 1550 nm windows; the chosen wavelength impacts the device cost, modulation bandwidth, power density, and atmospheric attenuation [72]. The optical beam is then collimated and steered toward the receiver aperture. At

the receiver, a telescope collects and focuses the incoming light through spectral and spatial filters to suppress background noise. A photodetector, typically a PIN diode or APD, converts the optical signal into an electrical current, which is then amplified, filtered, and processed to recover the transmitted data. Unlike conventional Gaussian-noise channels, the FSO channel carries optical power, imposing two key constraints: the instantaneous signal must be non-negative and its average power bounded by eye safety and system limits. Moreover, increasing the active area of the photodetector can improve SNR but also increase the capacitance of the device, thus reducing the electrical bandwidth [73]. Fig. 2 shows the generic block diagram of an FSO system, comprising a transmitter, an atmospheric channel, and a receiver whose distinct design considerations and components are detailed in this section.

In practical FSO communication, noise is an inherent part of the system, either from temperature variations, air turbulence, or even from the detectors themselves. The errors resulting from these disturbances make the retrieval of transmitted information a very challenging task [74]. Three essential resources that system designers can control in order to combat this are bandwidth, power, and time. Bandwidth B in Hertz refers to the range of frequencies available to transmit a signal. A larger bandwidth allows for a higher potential data rate. More data can be transmitted simultaneously with more bandwidth, although it may be constrained by power or complexity. A higher signal-to-noise ratio (SNR) allows for reliable transmission by reducing the BER [50]. It affects the channel capacity C as well, which is described in the Shannon formula as [75]

$$C = B \log_2(1 + \text{SNR}). \quad (1)$$

Thus, capacity quantifies the maximum achievable data rate for a given BER, transmit power, and channel bandwidth under noisy conditions. In addition, increased transmission power improves signal strength to go through noise, but it consumes more energy and may require expensive equipment. While the term “time” describes how long it takes to transfer each piece of data. Better precision may be ensured by allowing a longer time; however, this slows down the transmission rate. These three resources are interconnected [76]. To compensate for any restricted availability of one, adjustments can be made to the other resources. For instance, with limited bandwidth availability, reliable transmission can be achieved by increasing signal power or slower rates. Thus, balancing these resources, sending as much data as possible with the least amount of bandwidth while keeping the BER performance, is an inevitable concern. Signal modulation and channel coding are fundamental to mitigating noise in FSO communication systems. Consequently, high data rates can be achieved with complex modulations but require robust coding to handle errors effectively [77].

B. Channel Modeling

One of the primary reasons for system reliability degradation is the random variation in signal irradiance caused by the atmospheric channel. Atmospheric channels consist of particles of various molecules of gases, fog, smoke, dust, rain, etc. that

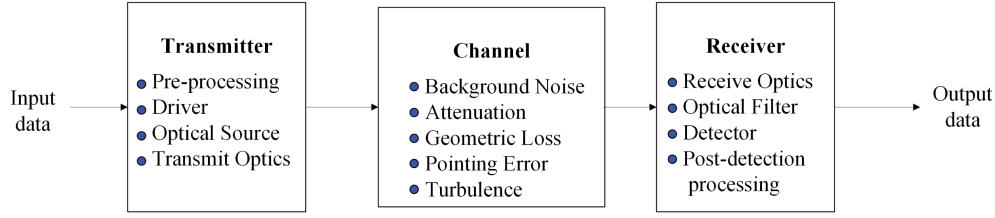


Fig. 2. Standard Block Diagram of FSO Communication Systems.

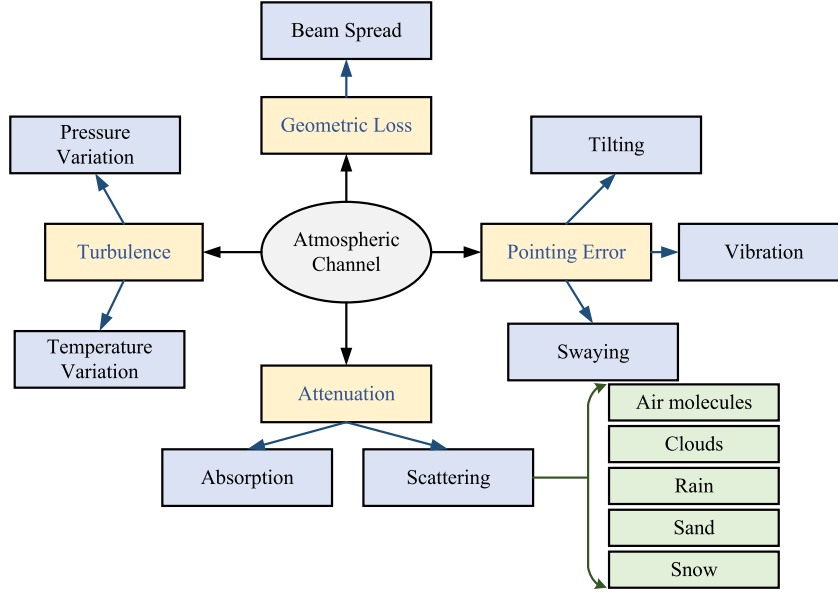


Fig. 3. FSO Atmospheric Channel Impairments.

affect the transmission, with the highest concentration in lower altitudes up to the ionosphere [78]. Fig. 3 illustrates the causes of atmospheric channel impairments, such as attenuation (I_{att}), turbulence (I_{tur}), pointing error (I_{pt}), and geometric loss (I_{geo}), where the combined state of the channel is

$$I = I_{att} I_{geo} I_{pt} I_{tur}. \quad (2)$$

Attenuation is the decrease in signal power while it propagates through the channel, and its main causes are scattering and absorption, which happen due to the existence of irregular particles. The overall attenuation in the received signal is deterministic and can be calculated from Beer's law as [79]

$$I_{att}(d) = \exp(-\gamma(\lambda)d), \quad (3)$$

where λ is the wavelength, γ is the attenuation coefficient, which is dependent on the absorption and scattering of molecules and aerosols, and d is the link distance. The choice of λ window is made to achieve minimum attenuation. Based on the size of different particles in the atmosphere relative to the wavelength of the light carrier, the scattering coefficient is determined. Also, there can be a total blockage of the line of sight (LoS) component when the obstacle diameter (D_o) is larger than the laser beam diameter (D_b) or a partial blockage when (D_o) is less than (D_b)

[80]. Table III summarizes how different types of scattering affect the FSO channel [81].

Another major impairment in FSO links is geometric loss, denoted I_{geo} , which arises from beam spreading during propagation. As the optical beam diverges, only a fraction of the transmitted power is captured by the receiver aperture, and the remainder is lost, reducing the received signal strength and degrading link performance [82]. Geometric loss can be modeled deterministically from the transmitter's beam divergence angle and the propagation distance; it is often approximated by the ratio of the receiver aperture area to the beam footprint area at the receiver plane. Using a transmitter with a narrow divergence angle mitigates these losses over long distances but increases sensitivity to misalignment: even slight pointing errors can cause significant power loss or total link outage.

I_{pt} arises from misalignment between the transmitted beam axis and the receiver aperture, which reduces the fraction of power actually collected. This misalignment results from factors such as platform vibrations, mechanical drift, thermal expansion or contraction, and wind-induced sway, all of which introduce random angular and lateral displacements. In most analyses, the instantaneous pointing offset is modeled as a zero-mean Gaussian random variable, leading to a Rayleigh (or bivariate

TABLE III
COMPARISON BETWEEN SCATTERING TYPES

Scattering Type	Particle Size	Particle Examples	Wavelength-Dependence	Impact on FSO Communication
Rayleigh	Much smaller than λ	Air molecules	Strongly	Scattering in all directions, negligible for larger λ
Mie	Comparable to λ	Haze, fog	Less	Severe attenuation
Non-selective	Much larger than λ	Sand, rain, clouds	Independent	Reduced visibility and signal loss

Gaussian) distribution for the radial displacement at the receiver plane [83]. Accurately characterizing I_{pt} is essential for predicting link reliability and designing precision PAT subsystems. Although present in all FSO systems, its severity and underlying causes differ substantially across terrestrial, aerial, and space environments because of differences in link distance, beam divergence, and platform motion.

Terrestrial links typically span a few hundred meters to several kilometers, with transceivers mounted on relatively stable structures such as rooftops or towers [84]. The dominant pointing-error sources are building sway, thermal expansion, and low-frequency mechanical vibrations. To simplify pointing acquisition and reduce sensitivity to minor misalignment, terrestrial systems often employ wider beam divergence, leading to a larger equivalent beam radius. Making terrestrial systems moderately tolerant to pointing errors. In many practical deployments, initial boresight offset is more problematic than dynamic jitter, and the performance can often be improved by careful mechanical alignment.

Aerial FSO links experience the most severe and dynamic pointing errors [85]. UAVs or high-HAPs operate in turbulent atmospheric layers and are subject to airframe vibration, engine-induced oscillations, wind gusts, and continuous attitude corrections. As a result, the jitter variance is typically much larger than in terrestrial links. Because aerial links span longer distances and require higher link budgets, the beam divergence must be kept narrow, indicating strong susceptibility to misalignment. Aerial FSO systems rely heavily on PAT, adaptive beam steering, and mechanical stabilization to maintain link reliability.

Space-based FSO links, encompassing ISLs, satellite-to-ground, and deep space links, present a unique set of pointing challenges governed by extreme distances and the orbital environment. For ISLs, the absence of atmospheric turbulence is a benefit, but the ultra-narrow beam divergence (on the order of microradians) to overcome path loss makes the system exceptionally sensitive to even minor perturbations [86]. In satellite-to-ground links, the pointing error model becomes more complex, as it must account for jitter from both the satellite platform and the ground station, the latter of which is also subjected to atmospheric turbulence-induced angle-of-arrival fluctuations. Deep space links push these challenges to the fundamental limits of current technology. With propagation distances spanning millions of kilometers, the beam footprint is enormous, and the received power is exceedingly weak [87]. To achieve a viable link budget, the transmitter must employ an exceptionally narrow beam, resulting in a beamwidth that can be narrower than the pointing uncertainty of the spacecraft by an order of magnitude or more. This necessitates a multi-stage pointing approach: a coarse pointing mechanism aligns the spacecraft,

while a fine pointing mechanism, often utilizing a beacon signal from Earth, provides sub-microradian stabilization [88].

Turbulence I_{tur} is caused by random fluctuations in the refractive index of the atmosphere C_n^2 , which occur from random changes in temperature, wind speed, and pressure [89]. The shape, direction, and electromagnetic characteristics of a laser beam are thus influenced by atmospheric turbulence. In a clear environment, the attenuation coefficient is considered small, and turbulence becomes the major source of impairment in short-range FSO communications. I_{tur} is also a random process, and there have been numerous studies to model it for different turbulence levels, as will be discussed later. Fading due to turbulence can lead to three phenomena in the transmitted laser beam: beam wander, beam spreading, and scintillation [90]. We can classify scintillation-based turbulence on the Rytov variance parameter σ_R^2 that is dependent on C_n^2 , wavenumber k , and link distance d . For weak turbulence ($\sigma_R^2 < 1$), the channel probability density function (PDF) can be modeled using the log-normal and Gamma Gamma (GG), Rayleigh, Rician, Nakagami-m distributions [91]. For moderate turbulence ($\sigma_R^2 \sim 1$), the GG model is the most popular model along with the double Weibull model [92]. While the GG, K, and negative exponential models are widely adopted for strong turbulence ($\sigma_R^2 > 1$). In saturated turbulence ($\sigma_R^2 \rightarrow \infty$), the negative exponential is mainly used along with the GG and K distributions [93], [94]. The Malaga $\mathcal{M}$ distribution is a generalized statistical model that combines various turbulence scenarios discussed before, ranging from weak to strong, and also can take into consideration the pointing error effect [95], [96]. Special cases of the $\mathcal{M}$ model, such as Rice-Nakagami, gamma, shadowed-Rician, K, exponential, log-normal, or Gamma-Rician models, are derived based on the values of the LOS component power, the amount of scattering power coupled to the LOS component, and the amount of fading [97]. The amount of turbulence in terrestrial, aerial, and satellite communication channels determines the best model to be chosen to mimic the experimental data. Table IV summarizes the PDFs of commonly used turbulent channel models in FSO communication under various turbulence conditions, where $I > 0$ denotes the received irradiance, $k_a(\cdot)$ is the modified Bessel function of the second kind of order a , and $\Gamma(\cdot)$ is the gamma function. The turbulence-induced fading (scintillation) in a clear environment may be minimized through aperture averaging, strong error control coding, and diversity approaches [98]. For aperture averaging, the receiver aperture must be much larger than the spatial coherence distance ρ_o of air turbulence. This criterion is not often satisfied in FSO since the spatial coherence distance is in the order of centimeters. Thus, coding must be strong enough to detect and repair both burst and random errors. This is mostly owing to the temporal coherence

TABLE IV
FSO TURBULENCE CHANNEL MODELS SUMMARY

Model	Turbulence Level	$f_{I_{\text{tur}}}$	Parameters
LN distribution	Weak	$\frac{1}{I\sqrt{2\pi\sigma_I^2}} \exp\left[-\frac{(\ln(I/I_0) + \frac{1}{2}\sigma_I^2)^2}{2\sigma_I^2}\right]$	σ_I^2 is the variance of the log irradiance; I_0 is the mean irradiance
Rayleigh distribution	Weak	$\frac{I}{\sigma_I^2} \exp\left(-\frac{I^2}{2\sigma_I^2}\right)$	σ_I^2 is the variance
Nakagami-m distribution	Weak	$\frac{2M^M}{\Gamma(M)\Omega^M} I^{2M-1} \exp\left(-\frac{MI^2}{2\Omega}\right)$	M is the shape parameter, $\Omega = E(I^2)$
GG distribution	All levels	$\frac{2\sqrt{a\beta}^{(a+\beta)}}{\Gamma(a)\Gamma(\beta)} I^{\frac{(a+\beta)}{2}-1} K_{a-\beta}(2\sqrt{a\beta}I)$	a, β are positive parameters for the effective number of scattering cells for large-scale and small-scale irradiance fluctuations, respectively
K distribution	Strong Saturated	$\frac{2}{\Gamma(\alpha)} I^{(\alpha-1)/2} \alpha^{(\alpha+1)/2} K_{\alpha-1}(2\sqrt{\alpha}I)$	α is the effective number of discrete scatterers
Negative exponential	Strong, Saturated	$\frac{I}{I_0} \exp\left(-\frac{I}{I_0}\right)$	I_0 is the mean irradiance
$\mathcal{M}$ distribution	All levels	$A \sum_{k=1}^{\beta} a_k I^{\frac{\alpha+k}{2}-1} K_{\alpha-k}\left(2\sqrt{\frac{\alpha\beta I}{\gamma\beta+\Omega}}\right)$	α is a positive parameter for the effective number of cells for large-scale of the large-scale scattering process; β is a parameter for the amount of fading; Ω is the LOS component power; γ is the independent scatter component power; Ω is the coherent contribution average power
Exponentiated Weibull	Moderate, Strong	$\frac{\alpha\beta}{\eta} \left(\frac{I}{\eta}\right)^{\beta-1} \exp\left[-\left(\frac{I}{\eta}\right)^{\beta}\right] \left(1 - \exp\left[-\left(\frac{I}{\eta}\right)^{\beta}\right]\right)^{\alpha-1}$	α is a positive shape parameter that depends on the receiver's aperture size; β is a positive shape parameter depending on the scintillation index; η is positive, depends on β , and related to the irradiance mean

time τ_o of air turbulence, which is significantly longer than the symbol duration T [89]. The spatial diversity approach may effectively compensate for turbulence-induced fading. It can also greatly reduce error probability while increasing the capacity of an FSO link. However, spatial diversity needs numerous transmit and receive apertures, which may not be compatible with the FSO transceiver size-weight and power (SWaP) criteria. To overcome such challenges, several proposed techniques in literature, including adaptive optics to mitigate the effects of weather conditions, employ higher power transmitters, implement diversity techniques, and optimize wavelength selection to minimize attenuation [99], [100]. Another approach is to utilize high-precision tracking mechanisms and to maintain alignment between transmitter and receiver. In addition, shielding optical components and the usage of radiation-hardened materials mitigate the effects of space background radiation.

Direct, reflected, and scattered sunlight and starlight introduce background radiation into FSO links, which degrades receiver sensitivity and can even trigger link outages under intense illumination. To mitigate this, receiver designs typically incorporate spatial and spectral filtering before photodetection, carefully optimize the aperture size, and employ optics to limit the field-of-view and reject off-axis light [12]. When background levels are low, the resulting noise may be neglected or modeled as a Poisson process [101]. However, in high-radiation environments such as space-based FSO, the aggregate background behaves more like a Gaussian random variable, requiring additional filtering and signal-processing techniques to ensure accurate detection.

The effect of pointing error is to be combined with turbulence and attenuation to get the aggregated channel model to account for the whole channel impairment effects I [102], [103], [104].

Therefore, with the aid of channel modeling, the received signal power P_R can be predicted from the transmit power P_T as

$$P_R = \eta I P_T + n, \quad (4)$$

where η is the efficiency, I is the combined coefficient describing the channel, and n is the zero-mean Gaussian noise caused by background radiations with variance σ_n^2 . Accurately characterizing the channel gain PDF $f_I(I)$ is essential for evaluating key performance metrics, such as the unconditional bit-error rate, ergodic capacity, and outage probability. Consequently, a great deal of research has been devoted to developing precise statistical models for different atmospheric conditions. In these composite models, I_{att} and I_{geo} are treated as deterministic components, whereas pointing I_{pt} and I_{tur} are modeled as random processes. Let $c = I_{\text{att}}I_{\text{geo}}$, so that $I = c I_{\text{pt}}I_{\text{tur}}$. By the PDF scaling rule

$$f_I(I) = \frac{1}{c} f_{I'}\left(\frac{I}{c}\right), \quad (5)$$

where $I' = I_{\text{pt}}I_{\text{tur}}$. For the product of two independent random variables

$$f_{I'}(I') = \int_0^\infty \frac{1}{x} f_{I_{\text{pt}}}(x) f_{I_{\text{tur}}}\left(\frac{I'}{x}\right) dx. \quad (6)$$

Hence, the composite channel PDF is

$$f_I(I) = \frac{1}{c} \int_0^\infty \frac{1}{x} f_{I_{\text{pt}}}(x) f_{I_{\text{tur}}}\left(\frac{I}{cx}\right) dx. \quad (7)$$

By combining these effects, researchers derive comprehensive channel models that accurately reflect a wide range of different operating environments [11], [50], [105], [106].

For terrestrial FSO links, the relatively short propagation distances make pointing errors and beam divergence less severe than in aerial or satellite links, and beam alignment is generally more manageable. However, LoS can still be obstructed by buildings or terrain, so atmospheric turbulence and attenuation often dominate performance degradation.

Aerial FSO systems mounted on UAVs or HAPs face rapidly changing environmental conditions due to altitude variations and weather dynamics. These links exhibit time-varying fading, geometric losses, and pointing errors, and they may suffer from angle-of-arrival (AoA) fluctuations that can steer the beam outside the receiver's field of view [107]. Unlike fixed ground links, UAV-based FSO therefore requires channel models that jointly capture pointing jitter and AoA variations. Recent studies [108], [109] derive composite models for UAV-FSO channels, enabling efficient optimization of transmitter and receiver parameters to maintain visibility with reduced computational complexity.

Satellite FSO communications encompass uplinks, downlinks, and ISLs, each with distinct channel characteristics. Uplinks traverse the densest turbulent layer near the Earth's surface, resulting in stronger scintillation, whereas downlinks spend most of their path in the vacuum of space and encounter turbulence only near the end of the link [110]. Atmospheric scattering and absorption further attenuate the beam [111]. At orbital distances, beam divergence governs geometric loss, but weather does not affect inter-satellite links. Pointing errors must be tightly controlled, since the narrow beam divergence leaves little margin for misalignment [112]. As a result, satellite FSO channel models differ significantly from those for terrestrial and aerial scenarios.

Since the FSO channel is severely affected by the operating system, choosing the correct channel model for error calculation is vital and has been studied extensively in research [113], [114]. Atmospheric turbulence level (weak, moderate, or strong), diversity techniques, and coding are key factors in deriving the PDF of the received signal [115], [116]. The channel models detailed previously highlight the critical need for robustness against errors. This necessitates the application of efficient modulation and channel coding schemes to protect the integrity of the transmitted data.

IV. MODULATION TECHNIQUES

To communicate information across an optical channel, numerous approaches exist to modulate data onto the laser beam [117]. Optical modulation can be classified as direct or external modulation. Direct modulation involves directly modulating the optical source without the need for an additional optical modulator, whereas external modulation involves changing an optical wave's properties using a different modulator [118]. The optical modulators can either modulate unguided waves (wireless) or guided waves using waveguide devices. When a certain signal parameter is being altered, optical modulation can further be classified based on the form it takes, such as phase modulation, frequency modulation, polarization modulation, amplitude modulation, spatial modulation, and diffraction modulation. Also, modulation schemes are traditionally classified as analog

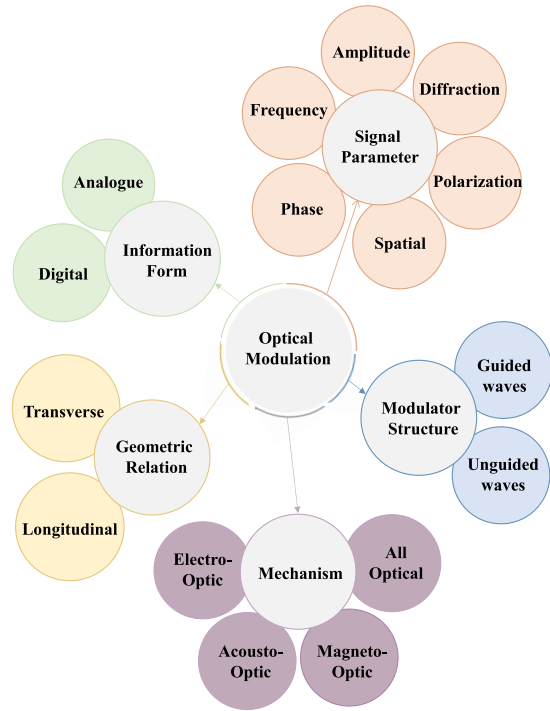


Fig. 4. Ways to Categorize Optical Modulation.

or digital based on their native signal format, though modern systems commonly transmit analog information using digital modulation techniques with appropriate conversion. Depending on the physical mechanism that causes the variation in optical susceptibility, optical modulation can be electro-optic, acousto-optic, magneto-optic, all-optical, and so on. Optical modulation can be transverse or longitudinal, based on the geometric relationship between the modulating signal and the modulated wave. Transverse modulation applies the signal perpendicularly to the optical wave's propagation direction, while longitudinal modulation applies the signal along the propagation direction of the optical wave. A special type of optical modulation is optical switching. It is the process of switching between two or more discrete values of an optical parameter or between two or more optical modes using large-signal digital optical modulation. We summarize the ways of categorizing optical modulation types in Fig. 4.

In FSO, the dominating signal-generating mechanism is to change the electrical signal input to the output optical power, thus electro-optic. Various modulation techniques can manipulate different field parameters, signal forms, generation mechanisms, and geometric relationships. As a result, a wide range of modulation techniques exists based on the combined schemes. The fundamental modulation schemes across all carrier frequencies include phase modulation (PM), frequency modulation (FM), and amplitude modulation (AM) for analog modulation. In digital modulation, these translate into phase-shift keying (PSK), frequency-shift keying (FSK), and amplitude-shift keying (ASK). With the help of digital signal processing hardware, digital signals can now be widely generated and easily adopted

in FSO systems. ON-OFF keying (OOK) and pulse amplitude modulation (PAM) are forms of ASK. It is implemented by mapping signal amplitude to two or more levels. PSK and FSK are alternatives for OOK and PPM; however, coherence detection is needed to detect the phase or frequency of the transmitted signal, which is more complex than direct detection. In the following subsections, different FSO modulation schemes will be discussed, and their main features and their suitability for each scenario.

A. Performance Metrics of FSO Modulation Schemes

Optimizing system performance based on the channel, atmospheric conditions, and application requirements is the motivation behind the development of various modulation techniques. To select the most suitable modulation technique for an FSO communication system, several critical metrics are to be evaluated, as follows:

1) *Spectral Efficiency*: Squeezing as much data into the least amount of spectrum feasible has become the primary objective of modulation schemes in recent times. Spectral efficiency, also known as bandwidth efficiency, is a measure of how effectively a given communication system utilizes its available bandwidth to transmit data. It represents the rate of information transfer per unit of bandwidth. Higher spectral efficiency means more data can be transmitted within a given bandwidth, making it a crucial metric in evaluating and comparing different modulation and coding schemes, especially in systems where bandwidth is a limited resource. The bandwidth efficiency, η_{BW} , defined as the ratio of the bit rate R_b to the required bandwidth B_{req} , is expressed as [119]

$$\eta_{BW} = \frac{R_b}{B_{req}}, \quad (8)$$

and is measured in bits per second per hertz (bps/Hz).

2) *Power Efficiency*: Power efficiency is how effectively a modulation scheme uses the available optical power to achieve reliable data transmission. It is measured by the required bit energy to noise power spectral density ratio (E_b/N_0) to achieve a BER. Optical power efficiency is measured by comparing the optical power gain of the modulation scheme P_{mod} over that of OOK (P_{OOK}) when both modulation schemes have the same minimum Euclidean distance d_{min} [120]. Approximately the average power requirement is

$$\eta_p = \frac{P_{mod}}{P_{OOK}}. \quad (9)$$

FSO transmission systems need to provide relatively power-efficient signals over long distances, such as in satellite or remote terrestrial applications, while also ensuring better receive sensitivity. This is crucial for maintaining reliable communication in the challenging environment of space, where power resources are limited, and signal attenuation can be significant. Effective FSO schemes must balance power efficiency, sensitivity, and robustness to ensure optimal performance in communication links. Power-efficient modulation schemes are easier to implement and effectively mitigate turbulence effects at low data rates. However, they must comply with eye safety regulations, which

limit the maximum propagation distance in highly turbulent conditions. The eye safety regulation is

$$\lim_{T \rightarrow \infty} \int_{-T}^T x(t) dt \leq P_{avg}, \quad (10)$$

where $x(t)$ is a positive signal and P_{avg} is the average optical power constraint for the light source.

Power-efficient modulation is crucial in any power-limited scenario, including channels with high turbulence levels and in deep-space communication. Whereas bandwidth-efficient modulation is favorable for commercial applications.

3) *Bit Error Rate*: For orthogonal L -ary slot-based modulations BER can be calculated from the symbol error rate (SER) depending on the modulation scheme. A symbol is encoded across L orthogonal slots, and a symbol error occurs if more than one slot has an error; thus, the probability that all slots are not in error is

$$P_c \approx (1 - \text{SIER})^L, \quad (11)$$

where SIER is the slot error probability and L is the average symbol length. So the SER is

$$\text{SER} \approx 1 - (1 - \text{SIER})^L. \quad (12)$$

BER can be expressed in terms of SER as [121]

$$\text{BER} \approx \frac{L/2}{L-1} \text{SER}. \quad (13)$$

In contrast, for non-slot-based modulations, the BER is derived from the SER using the constellation geometry and bit-to-symbol mapping.

The received signal intensity in FSO can change quickly due to atmospheric turbulence, resulting in severe signal deterioration over short periods of time. The conditional BER depends on the modulation scheme used and the SNR. The BER for a given instantaneous channel realization is referred to as the *conditional BER*. In contrast, the *unconditional BER* accounts for the statistical variation of the channel over time due to atmospheric turbulence. It represents the average BER computed over all possible channel states. The unconditional BER is obtained by averaging the conditional BER over the PDF of the channel irradiance $f_I(I)$. If the conditional BER in an additive white Gaussian noise (AWGN) channel is represented by $\text{BER}(I)$, then the average unconditional error probability is given by

$$P_e = \int_0^\infty \text{BER}(I) \cdot f_I(I) dI. \quad (14)$$

4) *Outage Probability*: In FSO communication, the outage probability describes the likelihood that the received SNR (γ) falls below a threshold (γ_{th}), which is required to maintain a desired quality of service (QoS). This metric is particularly important in the presence of atmospheric turbulence, which causes fluctuations in received power. The outage probability can be expressed as [122]:

$$P_{out}(\gamma_{th}) = P(\gamma < \gamma_{th}) = F_\gamma(\gamma_{th}) = \int_0^{\gamma_{th}} f_\gamma(\gamma) d\gamma, \quad (15)$$

TABLE V
OUTAGE PROBABILITY CHARACTERISTICS ACROSS FSO TURBULENCE REGIMES

Turbulence Regime	C_n^2 ($m^{-2/3}$)	Scintillation Index (σ_I^2)	Outage Floor Behavior	Applicable Channel Models [Refs]
Weak	10^{-17} – 10^{-15}	< 0.3	Negligible	Log-Normal, K-distribution [100], [126]
Moderate	10^{-15} – 10^{-14}	0.3 – 1	Low ($< 10^{-6}$)	Gamma-Gamma, K, $\mathcal{M}$ [125], [127]
Strong	10^{-14} – 10^{-13}	> 1	Moderate (10^{-4} – 10^{-5})	Gamma-Gamma, $\mathcal{M}$, Double-GG [128], [129]

where $F_\gamma(\cdot)$ and $f_\gamma(\cdot)$ denote the cumulative distribution function (CDF) and PDF of the SNR, respectively. The exact form of $f_\gamma(\gamma)$ depends on the channel model, pointing error statistics, and modulation scheme.

For mild atmospheric turbulence, the outage probability of a Gaussian FSO channel is proportional to $[\log(\text{SNR})]^2$. While under moderate to strong turbulence conditions, it is proportional to $\log(\text{SNR})$ [123]. The impact of atmospheric turbulence on outage probability varies significantly across different turbulence regimes. Table V summarizes the key characteristics of outage behavior under weak, moderate, strong, and very strong turbulence conditions, including the associated refractive index structure parameter (C_n^2), scintillation index σ_I , and applicable channel models. As turbulence intensity increases from weak to very strong conditions, the SNR penalty required to maintain a target outage probability escalates, while outage floor behavior transitions from negligible to significant. Moreover, the magnitude and emergence of outage floors depend on specific system parameters (beam width, receiver aperture, jitter variance, blockage probability and diversity) rather than turbulence intensity alone.

B. BaseBand (Pulse) Modulation

In FSO communication, binary-level formats are widely preferred for their simplicity and efficient use of power [14]. In OOK, the presence of a carrier pulse represents a binary ‘1’ and the absence represents a binary ‘0’. The average length (L_{avg}) is 1 b and its channel capacity is 1 as its bandwidth requirement $B_{\text{req}} = R_b$. OOK can use a return to zero (RZ) format in high-speed communication to improve robustness against inter-symbol interference (ISI), in which the pulse duration is a fraction of the whole bit duration, thus requiring more bandwidth [127]. When generally speaking about OOK, NRZ is the modulation referred to, in which the average transmitted power (power to transmit a ‘1’) $P_{\text{avg}} = P_t$ has a power requirement of 1. In IM/DD FSO systems, OOK-NRZ and OOK-RZ are not equivalent because the optical signal must remain non-negative and the noise statistics depend on instantaneous optical power. When both formats are compared under equal average power, the shorter RZ pulse requires twice the peak intensity of NRZ, resulting in a 3-dB SNR advantage. Furthermore, the reduced pulse width of RZ broadens its optical spectrum, leading to lower bandwidth efficiency relative to NRZ. And their BER is

$$\text{BER}_{\text{OOK-NRZ}} = \frac{1}{2} \text{erfc} \left(\frac{1}{2\sqrt{2}} \cdot \sqrt{\text{SNR}} \right), \quad (16)$$

$$\text{BER}_{\text{OOK-RZ}} = \frac{1}{2} \text{erfc} \left(\frac{1}{2} \cdot \sqrt{\text{SNR}} \right). \quad (17)$$

However, one of the primary issues with OOK (2-PAM), in atmospheric turbulence, is its sub-optimal performance, thus the requirement for adaptive thresholding. It is also less power efficient than other modulation techniques, as will be shown later.

For more than the two-level amplitude modulation of the OOK, PAM modulates 2^K bit signal into L possible amplitudes. Its BER is

$$\text{BER}_{L\text{-PAM}} = \frac{L-1}{L} \text{erfc} \left(\sqrt{\frac{3 \text{SNR}}{2(L^2-1)}} \right). \quad (18)$$

And the required bandwidth is

$$B_{\text{req}, L\text{-PAM}} = \frac{R_b}{\log_2 L}. \quad (19)$$

In comparison to a traditional OOK and other modulation schemes in FSO systems, L -PAM modulation decreases power efficiency and has the worst BER performance for higher L . That’s why it is not widely used for commercial and power-restricted applications.

Pulse position modulation (L -PPM) [48], however, encodes data by varying the position of the pulse within a fixed time period or frame, thus $L_{\text{avg}} = L = 2^K$, where K is the PPM order. The channel capacity for an ideal SNR is calculated from

$$C_{L\text{-PPM}} = K = \log_2(L). \quad (20)$$

And the required bandwidth and bandwidth efficiency are

$$B_{\text{req}} = \frac{LR_b}{\log_2(L)}. \quad (21)$$

$$\eta_{\text{BW}} = \frac{\log_2(L)}{L}. \quad (22)$$

In AWGN, the conditional BER performance of PPM can be calculated from

$$\text{BER}_{\text{PPM}} = \frac{1}{2} \text{erfc} \left(\frac{1}{2\sqrt{2}} \sqrt{\text{SNR} \frac{L}{2 \log_2 L}} \right). \quad (23)$$

To decrease the BER of PPM, L should be increased, which will enhance power efficiency in turn, but at the cost of reduced bandwidth efficiency. Therefore, to obtain power and bandwidth efficiency, modification is made to PPM.

Differential Pulse Position Modulation (DPPM) is a variable-length PPM format where the symbol duration is determined by the position of the pulse relative to the previous pulse, rather

TABLE VI
 MAPPING OF 3-BIT OOK INTO PPM, DPPM AND DPIM SYMBOLS

Scheme	3-bit OOK-NRZ	3-bit OOK-RZ	8-PPM	8-DPPM	8-DPIM (NGS)	8-DPIM (1GS)
	000	000	10000000	1	1	10
	001	001	01000000	01	10	100
	010	010	00100000	001	100	1000
	011	011	00010000	0001	1000	10000
	100	100	00001000	00001	10000	100000
	101	101	00000100	000001	100000	1000000
	110	110	00000010	0000001	1000000	10000000
	111	111	00000001	00000001	10000000	100000000
L_{avg}	3	3	8	9/2	9/2	11/2
B_{req} / R_b	1	2	8/3	3/2	11/6	11/3

than fixed time slots [128]. Each pulse immediately terminates the current symbol and starts the next, providing inherent symbol synchronization without the need for external timing recovery, unlike PPM. Another advantage of this adjustment is that it reduces the average symbol duration, thereby improving overall bandwidth efficiency. Its average length is

$$L_{avg,DPPM} = \frac{L+1}{2}. \quad (24)$$

And its capacity is

$$C_{DPPM} = \frac{2L \log_2(L)}{L+1}. \quad (25)$$

The required bandwidth and bandwidth efficiency are

$$B_{req,DPPM} = \frac{(L+1)R_b}{2 \log_2(L)} \quad (26)$$

$$\eta_{BW,DPPM} = \frac{2 \log_2(L)}{L+1}. \quad (27)$$

Another PPM variant is digital pulse interval modulation (DPIM), based on the interval between consecutive pulses relative to a reference interval, data is encoded [129]. Where each symbol starts with a pulse and is followed by a number of empty guard slots (GS) based on the decimal value of the data being encoded. DPIM symbols with no guard slots (NGS) are the mirror image of DPPM. DPIM only requires chip synchronization and no symbol synchronization is needed because each pulse initiates a symbol [130]. Its average length depends on L and the guard slots GS and is calculated from

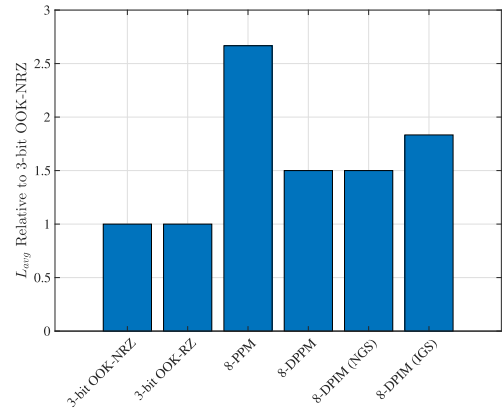
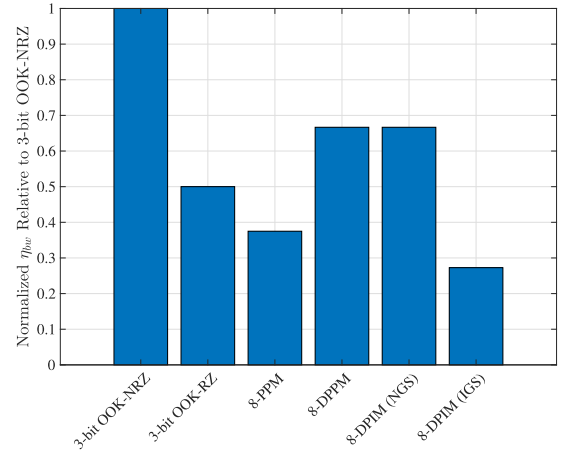
$$L_{avg,DPIM} = \frac{L+2GS+1}{2}. \quad (28)$$

Its capacity, required bandwidth, and bandwidth efficiency are

$$C_{DPIM} = \frac{2L \log_2(L)}{L+2GS+1}. \quad (29)$$

$$B_{req,DPIM} = \frac{(L+2GS+1)R_b}{2 \log_2(L)}. \quad (30)$$

$$\eta_{BW,DPIM} = \frac{2 \log_2(L)}{(L+2GS+1)}. \quad (31)$$


 Fig. 5. Average Length L_{avg} Relative to 3-bit OOK.

 Fig. 6. Normalized Bandwidth Efficiency η_{bw}/R_b Relative to 3-bit OOK.

For DPIM and DPPM, they are a varying length modulation scheme where information is decoded in the intervals between pulses rather than individual bits. Typically SER is used rather than BER in such schemes. Several studies have investigated the performance of SER in FSO and derived expressions for it [131], [132], [133]. To illustrate, Table VI provides a comparison using symbol mapping for 3-bit OOK, 8-PPM, 8-DPPM, and 8-DPIM modulation schemes. Fig. 5 and 6 give insights on the increase in

symbols length and enhancement/degradation in the bandwidth efficiencies of PPM, DPPM, DPIM with no guard slot, and DPIM with one guard slot with respect to the 3-bit OOK symbol.

In addition, some research proposed modulations based on PPM and DPPM having multi-amplitude versions called pulse amplitude and position modulation (PAPM) and differential amplitude pulse position modulation (DAPPM), respectively [134], [135]. PAPM and DAPPM are schemes with varying lengths depending on the amplitude levels (L_A) in the multi-level scheme. For the PAPM, L_{avg} is

$$L_{avg,PAPM} = \frac{L}{L_A}. \quad (32)$$

While for DAPPM it is

$$L_{avg,DAPPM} = \frac{L + L_A}{2L_A}. \quad (33)$$

Extensive research was conducted to compare the performances of different pulse modulation schemes in FSO applications. In [128], they compared OOK, PPM, and DPPM, taking into consideration AT, pointing error, and weather conditions. Although DPPM offers more capacity, PPM has the best BER performance among all schemes. The performance of the L -PPM system further improves with increasing the order. The performance of PPM was compared in [136] to OOK, DPPM, DAPM, and DAPPM. PPM performs best among the rest for both uplink and downlink. In the case of uplink, more bandwidth-efficient techniques can be employed at the expense of marginally higher transmitted power because PPM has the biggest bandwidth requirement. Combining PPM and OOK with turbo coding was implemented in [60], and PPM with coding is superior to both OOK with coding and PPM without coding. In reference to OOK, the advantages of PPM and DPIM are examined in [137]. With regard to the turbulence-induced intensity variation in the ground-to-satellite laser link, the packet error rates of these three modulation techniques are evaluated for both uplink and downlink. OOK-DD systems with optically preamplified receivers typically have a sensitivity of 70-80 ppb at a BER of 10^{-9} . Though OOK is simpler, PPM, on the other hand, doesn't require adaptive thresholding and is power efficient, but it is not bandwidth efficient and requires complex synchronization between the transmitter and receiver when there is jitter. Consequently, PPM emerges as the preferred choice for applications in deep space communication, as it is a photon-starved environment.

C. Passband Modulation

Using multiple subcarrier intensity modulation (SIM) in FSO systems enables the use of advanced modulation formats such as M -ary phase-shift keying (PSK), where information is conveyed through the carrier phase. The simplest case is binary PSK (BPSK), which employs two phase states separated by 180° . Unlike OOK and PAM, BPSK does not require adaptive thresholding in turbulence-impaired IM/DD channels [119]. Its conditional BER under coherent detection is

$$\text{BER}_{\text{BPSK}} = \frac{1}{2} \text{erfc}\left(\sqrt{\text{SNR}}\right), \quad (34)$$

and it requires a bandwidth of $B_{\text{req}} = R_b$. To improve spectral efficiency, BPSK can be generalized to M -PSK, where each symbol represents $\log_2 M$ bits. Increasing M enhances the bandwidth efficiency by $\log_2 M$, but the angular separation between adjacent constellation points decreases according to

$$\Delta\theta = \frac{2\pi}{M},$$

which reduces the noise margin and degrades BER performance. Among these formats, quadrature PSK (QPSK, $M = 4$) is particularly attractive for FSO because it doubles the bandwidth efficiency ($B_{\text{req}} = 0.5R_b$) while maintaining moderate sensitivity. Its BER under coherent detection is

$$\text{BER}_{\text{QPSK}} = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{\text{SNR}}{2}}\right). \quad (35)$$

However, coherent M -PSK systems require accurate carrier phase recovery, and the complexity of synchronization and phase-tracking circuitry increases with M .

Differential PSK (DPSK) is a non-coherent modulation scheme in which information is encoded in the phase difference between successive symbols, eliminating the need for explicit carrier phase recovery at the receiver [76]. In binary DPSK (DBPSK), bits are represented using phase shifts of 0° or 180° relative to the previous symbol. Differential QPSK (DQPSK) extends this concept to two bits per symbol by employing four possible phase transitions ($0^\circ, 90^\circ, 180^\circ, 270^\circ$), improving spectral efficiency while maintaining robustness to phase noise and phase ambiguities caused by atmospheric turbulence. Although DPSK receivers are simpler than coherent PSK receivers because no carrier phase tracking is required, they incur an inherent 3 dB SNR penalty in BER performance. The conditional BER of DBPSK in AWGN is

$$\text{BER}_{\text{DBPSK}} = \frac{1}{2} e^{-\text{SNR}}. \quad (36)$$

Quadrature amplitude modulation (QAM) is a modulation scheme on both carrier amplitude and phase dimensions [138]. L different amplitude and phase combinations are to form a symbol, thus increasing rates without requiring more bandwidth. However, to maintain the BER performance, more SNR is needed at the expense of power efficiency [3]. Finding a closed expression for SER in high-order L -QAM is challenging. Research is ongoing to find accurate upper and lower bounds for error performance in FSO atmospheric channels [139], [140], [141]. An approximation for the BER of square, gray-coded L -QAM [142] is

$$\text{BER}_{L\text{-QAM}} \approx \frac{2\left(1 - \frac{1}{\sqrt{L}}\right)}{\log_2(L)} \text{erfc}\left(\sqrt{\frac{3\text{SNR}}{2(L-1)}}\right). \quad (37)$$

M -ary FSK is a less used modulation due to its complexity in FSO, where the signal is encoded in M carrier frequencies [76]. M -FSK has $\log_2 M$ expansion in B_{req} and requires M frequency decisions at the receiver to recover the signal. That is why it is a less attractive scheme though it has good BER performance.

Without requiring additional power or bandwidth, the rate of M -ary schemes can be increased further by utilizing polarization shift keying (PolSK) [76]. PolSK modulates the signal by changing the light polarization states, making it a power and bandwidth-efficient choice.

FSO research continues to explore modulation schemes that balance data rate, robustness to channel turbulence, and energy efficiency. Using DPSK for heterodyne detection in [143] is considered since it achieves a 3 dB sensitivity advantage over OOK and lower peak power, which reduces any nonlinear effects. They also deployed numerous apertures at the satellite to reach an acceptable error rate target of 10^{-6} . While DPSK provides a 3dB increase over OOK, PolSK provides a 3.2dB advantage over OOK only in peak-power-limited systems and with balanced detection [144]. PolSK provides superior outage probability performance, whereas DPSK modulation styles achieve high channel capacity. Not all receiving techniques are appropriate for each modulation format. DD receivers are insensitive to phase and polarization information and require conversion into OOK using additional optical components, such as delay demodulators for DPSK or polarization filters for PolSK [64].

Combining baseband with passband modulation, considering the specific advantages of each modulation, hybrid modulation is formed. SIM that was borrowed from conventional RF communication underwent extensive research thereafter [145] in FSO. SIM doesn't require adaptive thresholding nor as much bandwidth as PPM, but it suffers from high PAPR because the source is on for both '0' and '1', having sinusoidal subcarriers that have both negative and positive values, thus requiring DC bias. SIM is already successfully deployed and mature in RF applications and optical fiber communications. It increases capacity by modulating data from different users on different subcarriers. However, it has relatively high transmitted power and is subject to distortion due to laser nonlinearity and clipping. It has been demonstrated that hybrid modulation techniques perform well in environments characterized by higher atmospheric turbulence. Conversely, in conditions of lower turbulence, such as communication between satellites or from satellites to the moon, single modulation methods are considered more effective. Examples of hybrid modulations include PPM-BPSK, PPM-MSK, PPM-GMSK, PPM-BPSK-SIM, PPM-FSK, PPM-OFDM, PPM-POLSK-SIM, and others [146], [147], [148]. A hybrid modulation scheme known as PPM-FSK-SIM is proposed in [149]. It utilizes PPM, MFSK, and SIM techniques. PPM-MFSK-SIM outperforms PPM and minimum shift keying (MSK) in terms of error rate in the presence of AT. In [150], they studied the performance of an Earth-to-HAP FSO communication system using OOK and SIMO to overcome the effects of atmospheric turbulence, pointing errors, and AoA fluctuations. When fast fading is not a significant problem in HAPs, coherent demodulation may be used. In [151], they utilized a sub-carrier intensity modulation approach combined with MPSK-SIM for FSO communication, deploying multiple HAPs to enhance the performance. In [25], a closed-form formula for average BER, channel capacity, and outage probability for several modulation schemes, including BPSK-SIM, DPSK, DPSK-SIM, M -ary PPM, and PolSK. They evaluated and compared the FSO system's channel capacity,

BER, and outage performance in strong atmospheric turbulence with the combined effect of turbulence and pointing errors.

D. Multi-Carrier Modulation

Multi-carrier modulation (MCM) is the process of independently modulating several digital data streams onto independent subcarriers, each operating at a different frequency [152]. MCM addresses the varying levels of attenuation and delays the signal experiences by splitting the overall signal into multiple narrowband subcarriers. OFDM, used in many RF and wired applications, attracted FSO communication attention as well. Each subcarrier is encoded using conventional multilevel passband modulation methods like M -PSK or L -QAM [153]. In this manner, compact spacing of the orthogonal subcarriers increases the data rate and is robust to ISI, thus resulting in better BER performance. Nonetheless, these subcarriers have both positive and negative values, which will be clipped by the optical source. Thus, adding DC will decrease the power efficiency of such schemes [154]. It also suffers from a high PAPR due to the dynamic range of the term summation, which severely impacts IM/DD FSO links. The nonlinear characteristics of practical optical sources cause clipping of OFDM signal peaks, leading to in-band distortion and BER degradation exceeding one order of magnitude [155]. This challenge is particularly acute in FSO due to the interaction between high PAPR and intensity modulation constraints. Mitigation techniques, including digital predistortion, PAPR reduction (selective mapping, clipping), and adaptive biasing, when combined with DCO-OFDM or ACO-OFDM variants, can suppress nonlinear distortion [156], making OFDM viable for FSO under proper transmitter design. Several work was dedicated to exploring OFDM-FSO systems error rates in atmospheric turbulence, and many have combined other techniques such as polarization division multiplexing (PDM) and FEC [157], [158], [159], [160]. However, the quest for a higher data rate comes at the expense of shorter ranges. In [161], it is shown that BPSK-OFDM performs better when compared to QAM-OFDM in a turbulence FSO channel. [162] showed that QAM-OFDM outperforms PSK-OFDM. A cascaded Erbium doped fiber amplifier (EDFA) was proposed in [60] to achieve a 30 Gbps rate at accepted error performance for these distances: 35 km, 3.9 km, 1.48 km, and 0.68 km, depending on the turbulence level. An 80 Gbps system utilizing wavelength-division multiplexed-mode-division multiplexed (WDM-MDM) OFDM is implemented in [163] for 15, 2.5, 1.8, and 1.1 km ranges in different weather conditions. OFDM and PDM are combined with another multiplexing technique named optical code division multiple access (OCDMA) to further increase the system rate to 120 Gbps at a maximum range of 5 km [164]. 2-dimensional coding with OCDMA-OFDM achieved a rate of 1.5 Tbps in [165] to serve a huge simultaneous number of users with a power and spectral-efficient system. A 1.6 Tbps was reported in [166] that proposed a WDM-PDM-CO-OFDM for a maximum range of 20 km for clear conditions, which is reduced with increasing weather conditions.

The second most favorable MCM in FSO is discrete multitone modulation (DMT). It is an OFDM variant where subcarriers

with higher SNR are assigned more bits, unlike OFDM, in which all carriers have the same number of bits [167]. Typically, DMT has higher power efficiency than OFDM. This is due to DMT's ability to dynamically distribute power across frequencies according to their respective channel conditions, which may lessen the effect of high PAPR. However, it requires more algorithms to make allocations and adjustments in real time. 64-QAM DMT was implemented in [70] for a short range FSO system. Simultaneous transmission of DMT and NRZ-OOK signals employing derivative-subtractive sampling to separate the mixed signals and sparse compressive sampling to cut the DMT signal length was implemented in [168]. Within a 10-MHz bandwidth, this configuration yields a transmission throughput of 37.6 Mbps. Whereas [28] investigated the use of coding with 32-QAM DMT for turbulent channels. 16-QAM-DMT is sent over a 100 m single-mode fiber (SMF), 100 m FSO link, then a 4 m wireless link in [169]. Furthermore, a directly modulated quantum cascade laser (DM-QCL) was used to study the BER and stability of DMT in [170].

Evaluating the performance of different modulation schemes in FSO, as it depends heavily on the specific application and environmental factors, is essential. In summary, for OOK, BPSK, and DPSK modulation schemes, the required bandwidth equals the bit rate, resulting in a theoretical bandwidth efficiency of one bit per second per hertz 1 bps/Hz [171]. QPSK theoretically achieves a bandwidth efficiency of 2 bps/Hz because its required bandwidth is half its bit rate. However, practical bandwidth efficiency for QPSK typically ranges from 1.4 to 1.6 bps/Hz. RZ-OOK requires double the bandwidth compared to NRZ-OOK, resulting in a theoretical bandwidth efficiency of 0.5 bps/Hz. For L -PAM, as the number of bits increases, its bandwidth efficiency also increases. In contrast, the bandwidth efficiency for L -PPM decreases as the number of bits increases, making it less bandwidth efficient than other modulation techniques except for RZ-OOK when $L = 2$ or 4. PAM achieves higher bandwidth efficiency than all the previously mentioned modulation schemes, making it a bandwidth-efficient technique suitable for FSO communication systems if power limitation is not an issue. The L -PPM scheme is commonly used because of its high PAPR, which increases the power efficiency. BPSK and QPSK are more power-efficient modulation schemes compared to 4-PPM and 2-PPM, respectively. However, they are less power-efficient than 8-PPM and 16-PPM [119]. Also, DPSK has gained considerable attention for its power efficiency and its 3dB improvement over OOK modulation [172]. In addition to the modulation scheme, aperture averaging is crucial to enhancing the power efficiency of the FSO link [85]. Whereas in hybrid SIM modulations, their low optical power efficiency is a main disadvantage, which results from the necessity of adding a DC bias to the multi-subcarrier electrical signal before optical intensity modulation to prevent negative amplitudes [173]. MCM has the best performance in ISI, turbulent channels, and huge data rates are however advised in short-range communications (terrestrial).

Table VII summarizes the conditional BER expressions for the modulation schemes considered, and Fig. 7 illustrates their comparative performance. Key modulations require

TABLE VII
THE CONDITIONAL PROBABILITY OF ERROR FOR OPTICAL MODULATION EXPRESSED AS A FUNCTION OF SNR

Scheme	BER
OOK _{NRZ}	$\frac{1}{2} \operatorname{erfc}\left(\frac{1}{2\sqrt{2}} \cdot \sqrt{\text{SNR}}\right)$
OOK _{RZ}	$\frac{1}{2} \operatorname{erfc}\left(\frac{1}{2} \cdot \sqrt{\text{SNR}}\right)$
L -PPM	$\frac{1}{2} \operatorname{erfc}\left(\frac{1}{2\sqrt{2}} \cdot \sqrt{\text{SNR} \cdot \frac{L}{2} \log_2 L}\right)$
L -PAM	$\frac{L-1}{L} \operatorname{erfc}\left(\sqrt{\frac{3 \text{SNR}}{2(L^2-1)}}\right)$
BPSK	$\frac{1}{2} \operatorname{erfc}(\sqrt{\text{SNR}})$
DBPSK	$\frac{1}{2} e^{-\text{SNR}}$
QPSK	$\frac{1}{2} \operatorname{erfc}(\sqrt{\text{SNR}/2})$
L -QAM	$\frac{2}{\log_2(L)} \left(1 - \frac{1}{\sqrt{L}}\right) \operatorname{erfc}\left(\sqrt{\frac{3 \text{SNR}}{2(L-1)}}\right)$

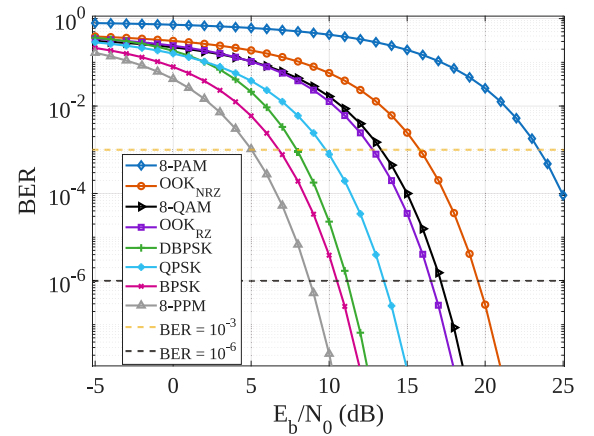


Fig. 7. BER vs. E_b/N_0 for Different Optical Modulation Schemes showing 10^{-3} , 10^{-6} threshold.

dramatically different E_b/N_0 , as figure 7 reveals a 10.9 dB performance gap between optimal 8-PPM (8.92 dB) and basic OOK-NRZ (19.82 dB) at $\text{BER}=10^{-6}$, with coherent PSK variants (BPSK: 10.80 dB, QPSK: 13.80 dB) providing intermediate efficiency. Phase-based modulations (BPSK, QPSK, and L -QAM) are evaluated under coherent detection, whereas intensity-based schemes (OOK, PAM, and PPM) employ IM/DD. All expressions correspond to conditional AWGN BERs at a fixed SNR, without averaging over atmospheric turbulence. Subsequently, other modulation schemes' error performance, such as PPM variants, is evaluated through SER or packet error rate (PER) rather than BER [136], [174]. For MCM, the BER expression is derived based on the passband modulation scheme used.

System complexity describes the degree of technical difficulty and circuit integration in a system's design. The manufacturing costs, which are a major factor in choosing a modulation technique, are related to the complexity. Generally speaking, modulators are simpler than demodulators. In particular, coherent demodulators are more complex than noncoherent demodulators because they need carrier recovery. The system's complexity

TABLE VIII
COMPARATIVE ANALYSIS OF FSO MODULATION FORMATS

Modulation Scheme	Bandwidth Efficiency η_{BW} (bps/Hz)	Power Efficiency η_p (vs. OOK NRZ) (dB)	Receiver Complexity	Recommended Application
OOK-NRZ	1.0	0 (Ref.)	Low	Short-range terrestrial
OOK-RZ	0.5	+1.5	Low	High-speed terrestrial
8-PPM	0.375	+5.4	Medium-High	Satellites, deep space (low data rate)
16-PPM	0.25	+7.5	High	Deep space
4-PAM	2.0	-3.3	Medium	Terrestrial short-range
BPSK	1.0	+3.5	Medium	Coherent satellite links
DPSK	1.0	+3.0	Medium	Aerial/UAV systems
QPSK	2.0	+2.8	Medium-High	Satellite
8-PSK	3.0	-6.1	High	Bandwidth-critical aerial
16-QAM	4.0	-8.75	High	Metro FSO, very low scintillation
OFDM-QPSK	2.0	+2.8 + PAPR penalty	Very High	Terrestrial
OFDM-16QAM	4.0	-8.75+ PAPR penalty	Very High	High-rate terrestrial
PolSK	1.0	+3.0	Medium-High	Satellite

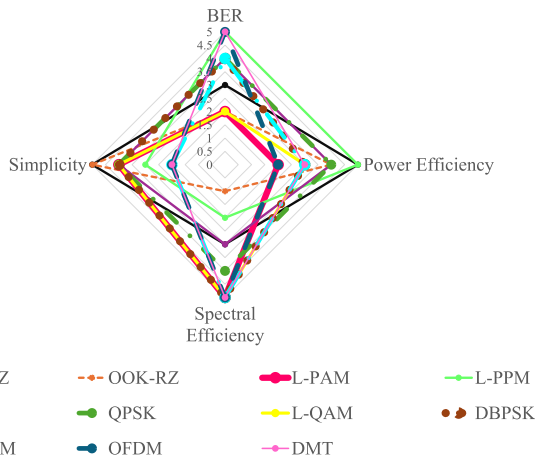


Fig. 8. Comparative Analysis of FSO Modulation Formats.

is further increased by the possibility that some demodulation techniques require sophisticated algorithms. Therefore, when evaluating the difficulty of various modulation techniques, all of these characteristics are crucial to be taken into account.

When choosing a modulation scheme for an FSO system, it's important to consider the specific needs and constraints of the application. For power-limited scenarios, prioritizing schemes that are power-efficient, like PPM, DPSK, or BPSK, is key. In bandwidth-constrained such as commercial applications, choose bandwidth-efficient schemes such as QPSK, *L*-QAM, or SIM. If cost and complexity are concerns, select simpler schemes like OOK-NRZ or OOK-RZ to keep costs low and simplify implementation. Based on the previous discussion, the chart in Fig. 8 provides a comparative analysis between different

modulation schemes in FSO systems, highlighting key metrics such as spectral efficiency, power efficiency, complexity, and BER performance. Table VIII compares modulation formats (including spectral efficiency, power-efficiency, complexity, and best FSO suitable application); the normalized power and bandwidth efficiency metrics presented are representative values [175] and should be interpreted relative to specific turbulence conditions, as actual performance is highly dependent on atmospheric channel characteristics, link distance, and scintillation, as detailed in the corresponding referenced studies.

V. CHANNEL CODING

Channel coding, which is considered a form of time diversity, is essential to exploit the channel capacity while ensuring high-quality transmission [76]. In digital communication, errors in FSO are mainly due to impairments in the atmospheric channel. While increasing the SNR to maintain the BER is not always applicable, coding proposes a solution for not only detecting errors but also correcting them without the need for higher signal power. Lower SNR requirements can leverage the power constraints from the transmitter cost [154]. Depending on the communication link tolerance for delay and increased system complexity, temporal diversity offered by coding can minimize the turbulence effect and improve performance. Several coding schemes have been studied to overcome the FSO turbulence channel [176], [177]. The smaller the required signal power using coding, the better the coding performance. The reduction in power between using coding and no coding is the coding gain and is described as

$$\Gamma_{code} = 10 \log_{10} \frac{P_{Req}(uncoded)}{P_{Req}(coded)}. \quad (38)$$

The encoding process for FEC codes in FSO involves adding redundant data to the original message m to create a coded sequence c . This can be expressed as

$$c = \text{Encoder}(m). \quad (39)$$

The encoding function varies for different coding schemes depending on the number of extra bits added and how they are incorporated. Thus, coding efficiency as a measure of how the coding scheme utilizes the bandwidth is another important metric to evaluate the coding performance. Higher code efficiency means fewer redundant bits, therefore less transmission overhead and better bandwidth utilization. However, this comes at the coding correction capability cost. The code efficiency can be described as

$$\eta_{\text{code}} = \frac{k}{n}, \quad (40)$$

where k is the number of information bits and n is the total bits with redundancy. At the receiver, the decoding process to retrieve the original message after passing through the noisy channel is performed as

$$\hat{m} = \text{Decoder}(c + \text{noise}). \quad (41)$$

The decoding function is designed to reverse the encoding process to detect and correct errors in order to reconstruct the original message m . The utilization of various ECC methods, such as Turbo, convolutional, Reed-Solomon (RS), trellis-coded modulation (TCM), and Low-Density Parity-Check (LDPC), further enhances the performance of FSO links and will be discussed in the following section.

A. Channel Coding Techniques

Convolutional codes are sequential error-correcting codes in which the output bits are regenerated from current and previous input bits. The encoder continuously produces output bits with redundant bits for error detection and correction [178]. They provide efficient error correction using sequential coding and the Viterbi algorithm, which is ideal for real-time decoding in FSO applications. Convolutional coding enhances data reliability and reduces bit error rates in FSO systems, thus improving the overall link performance.

Turbo codes are made up of two or more convolutional codes concatenated in parallel or serial fashion and divided by means of interleavers [178], [179]. There are three possible configurations in which turbo codes can be made: serial concatenated convolutional codes, parallel concatenated convolutional codes, and hybrid concatenated convolutional codes. The basic and most favored configuration is when two or more systematic recursive convolutional component encoders are connected by an interleaver to form parallel concatenated convolutional codes [180]. Comparing this structure to typical codes, large coding gains and performance increases are possible. Data transmission over FSO networks using turbo codes is more reliable thanks to its iterative decoding process, which offers a significant improvement in error correction.

In FSO communication systems, RS codes are frequently employed for error correction, particularly over strong turbulent channels. These are cyclic error-correcting codes that are non-binary and work with symbols instead of bits [181]. These algorithms can identify more errors and correct a certain number of them. They are effective at correcting errors that arise in data blocks, which makes them appropriate for situations where errors often burst, as in strong-turbulent air conditions.

LDPC codes are characterized by their sparse parity-check matrices. This sparsity enables the use of low-complexity, iterative message-passing decoding algorithms, which approach the channel capacity. While the decoding complexity of LDPC codes can be higher than traditional block codes such as Reed-Solomon, this complexity varies with code size and structure [182]. LDPC codes offer superior efficiency and error correction, especially at high data rates, making them ideal for high-speed FSO systems where robust error correction is essential to maintain reliable communication despite atmospheric turbulence and other channel impairments.

A particular type of FEC code is called a rateless code also referred to as a fountain code [183]. Rateless codes can adaptively adjust to varying channel conditions without needing to know the error rate in advance. This is beneficial for FSO systems where atmospheric turbulence can cause rapid changes in signal quality. Rateless codes make the system robust to packet loss by producing an endless stream of encoded symbols that enable the receiver to reconstruct the original data from any large subset of received symbols.

RS and convolutional codes provide sufficient coding gain under weak atmospheric turbulence. For a direct detection FSO system in [184] using PPM with perfect interleaving, a convolutional code with a constraint length of 3 and a code rate of 1/2 offers a maximum coding gain of 7 dB in clear weather and 11 dB in moderate turbulence. Turbo, Trellis, or LDPC codes are, on the other hand recommended in intense air turbulence [185]. Because LDPC codes have a lower decoding complexity and computational overhead than Turbo codes, they are recommended for very high data rate transmission. Additionally, boosting channel capacity and offering significant coding gain are possible with variable rate LDPC codes [186]. In [187], they proposed a concatenated RS-LDPC code with which performance is compared with rateless codes. For that water to air environment using a UAV receiver, rateless code results outperform that of the proposed RS-LDPC code.

There has been extensive research done over the years on error performance in fading channels utilizing error correction codes in FSO communication systems. As a result, the receiver design becomes more complex because large memories are needed to store long data frames. While this enhances the coding gain of the FSO system, it also introduces delay latencies and increases system complexity. Certain coding methods incorporate a variety of space, frequency, and polarization in addition to time diversity. To further enhance system performance in the presence of fading, a number of space-time coding techniques have been introduced. In [188], research was done on a log-normal turbulent channel, it has been demonstrated that orthogonal space-time block coding (OSTBC) is inefficient for IM/DD FSO systems,

and repetition codes are better. Such claims were reexamined in [189], where the authors demonstrate that space-time coding is superior to repetition codes for a coherent FSO system with heterodyne detection across gamma-gamma channels.

Aside from physical layer coding, which mainly addresses data transmission over physical media, link layer research concentrating on controlling data transmission between neighboring network nodes is also taken into account in FSO [190]. In order to increase throughput, strengthen reliability, and optimize energy consumption in wireless systems operating over time-varying fading channels, hybrid automatic repeat request (HARQ) is borrowed from RF [191]. HARQ is superior to traditional coding such that it combines ECC with ARQ protocols to overcome their limitations. When errors are found, ARQ relies on the receiver to request retransmission of the frames from the transmitter. These retransmissions take time in long-distance communications like satellite communications, which can impose considerable delays and restrict throughput even if they guarantee reliable data transfer. On the contrary, FEC adds extra bits, enabling error correction at the receiver without needing retransmissions. However, FEC frequently needs a high level of redundancy that can result in throughput inefficiencies. Therefore, HARQ makes use of both techniques, allowing error correction and retransmission when the signal is fully corrupted. Thus, balancing reliability and efficiency [192].

Adaptive channel coding and adaptive coded modulation (ACM) are other channel fading mitigation techniques where power is regulated to achieve the target BER and ergodic capacity with minimal outages. FSO systems have slower channel fading, allowing for more accurate channel state information (CSI) estimation and feedback to the transmitter [193], [194]. Thus the FSO transmitter modifies its parameters such as transmit power, modulation level, and coding rate based on the current CSI. Adaptive transmission is optimal for a full duplex in that a limited bandwidth is dedicated to feedback operations in FSO links [195].

B. Performance Metrics of FSO Channel Coding

Although applying coding techniques in FSO leads to reaching the maximum channel capacity limits while also ensuring reliability, it is practically difficult to implement. Therefore, some code merits are to be met, such as code efficiency and code gain (rate) [196]. Various aspects significantly impact the choice and performance of error-correcting codes in different FSO communication systems.

- **Burst vs. random errors:** Different coding schemes vary in performance strength according to the type of errors in the FSO channel. Burst and random errors both exist in FSO channels where burst errors are due to atmospheric turbulence and signal misalignment, while random errors are caused by background noise. RS codes with convolutional codes and LDPC codes are particularly famous for correcting burst errors because their structure allows the correction of successive and long error blocks [197], [198]. RS codes are also used For random error correction, which is why it is widely used in many applications [199].

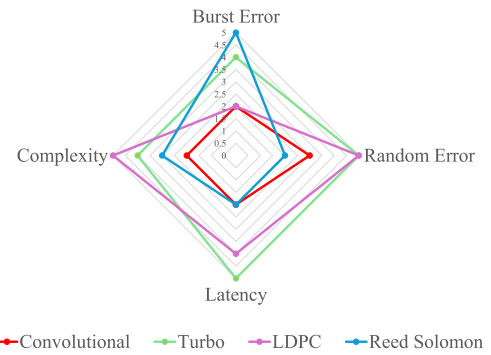


Fig. 9. Comparative Analysis of FSO Channel Codes.

Turbo codes and BCH codes are, on the other hand very appropriate for correcting random errors [200].

- **Latency:** For low-latency applications where real-time transmission is critical, codes with low delay during the coding and decoding process are preferable. LDPC and Turbo codes, although having strong error correction capability, introduce significant latency due to their iterative decoding processes [200]. Faster decoding speeds are achieved by simpler codes, such as convolutional and RS codes, but they may not be as effective in challenging circumstances.
- **Complexity:** Evaluating the complexity of a channel coding scheme depends on many aspects, such as processing and memory requirements [201]. Encoding complexity depends on the number of computations required to form a coded sequence, and the complexity increases with block length linearly or exponentially, depending on the code. Also, the error correction capability of a code can add to the decoding process complexity. In addition, codes relying on large matrices or look-up tables also add to the complexity aspect by requiring more memory for storage.

Based on the discussion, the chart in Fig. 9 provides a comparative analysis between the most used FSO channel coding schemes regarding their encoding and decoding complexity levels, latency, and how well they perform in random and burst errors. And Table IX summarizes channel coding schemes for FSO systems, showing code rate, coding gain, decoding complexity and latency, dominant error type, and recommended applications. Coding gain, however, is channel- and system-dependent, varying with modulation format, code parameters, SNR, and target BER/SER. In atmospheric FSO channels, random errors dominate under weak-to-moderate turbulence, while burst and combined random-burst errors are more significant under strong turbulence, as classified by the scintillation index σ^2 . The reported results provide useful reference-level insight into the performance of different coding schemes under their respective assumptions.

VI. TRADE-OFFS AND DESIGN GUIDELINES FOR MODULATION AND CODING IN FSO SYSTEMS

Across all FSO systems, the modulation and coding strategy must jointly consider turbulence robustness, pointing error

TABLE IX
COMPARATIVE ANALYSIS OF CHANNEL CODING SCHEMES IN FSO SYSTEMS

Coding Scheme	Code Rate	Coding Gain (dB) (vs. uncoded)	Decoding Complexity	Latency	Error Type Strength	Best Application	Ref.
Uncoded OOK	1.0	Baseline	—	—	Single errors	Terrestrial (very weak turb.)	[206]
BCH (255,131,19)	0.5	−2.0 (slow fading)	Moderate	Moderate	Burst errors (correlated)	Not suitable w/o interleaver	[206]
BCH Interleaver (255,131,19) +	0.5	22.8 (high turb.) 17.4 (mod. turb.) 4.5 (low turb.)	Moderate	Moderate	Burst errors dispersed	Aerial, Terrestrial (high turbulence)	[206]
Convolutional Code (3,1,2)	1/3	10.0 @ BER= 10^{-12}	Low	Low	Random errors (strong turbulence)	Terrestrial, Aerial (OFDM-FSO)	[210]
LDPC (2048,1024)	0.5	−2.3 (slow fading)	Moderate	High	Burst errors (correlated)	Not suitable w/o interleaver	[206]
LDPC (16935,13550)	0.80	9.0 @ BER= 10^{-8}	Moderate	High	Random errors	Terrestrial, Aerial (mod.-strong turb.)	[211]
LDPC Interleaver (2048,1024) +	0.5	25.1 (high turb.) 19.6 (mod. turb.) 5.4 (low turb.)	Moderate	High	Burst errors	Aerial, Terrestrial (high turbulence)	[206]
Nonbinary LDPC (16935,13548)	0.80	9.3 @ BER= 10^{-8}	High	Reduced latency	Burst errors	Terrestrial, Aerial (mod.-strong turb.)	[211]
Reed-Solomon (255,191,65)	0.75	13.0 @ BER= 10^{-5} 17.0 @ BER= 10^{-6}	Moderate	Moderate	Burst errors (with erasure)	Terrestrial, Aerial (weak-mod. turb.)	[212]
Reed-Solomon (255,127)	0.5	7.7 (clear, no turb.), 24-28 (strong turb.) @ BER= 10^{-3}	Moderate	Moderate	Burst errors	Terrestrial (moderate turb.)	[213]
Rateless Codes	Variable	10–15	Low	Low	Erasur errors	Satellite, Deep Space (very strong turb.)	[214]
Turbo Codes	0.5	3	Moderate	High	Random errors (strong turbulence)	Terrestrial, Aerial	[215]

tolerance, spectral and power efficiency, hardware limitations, and system complexity. Terrestrial systems emphasize turbulence mitigation, aerial systems require mobility compensation and SWaP efficiency, and space-based systems demand ultra-precise pointing and high reliability over long distances. The optimal design for each environment emerges from the interplay among these constraints, highlighting the need for environment-adaptive modulation, coding, and link control mechanisms tailored to the characteristics of each FSO scenario. The comparative analysis across terrestrial, aerial, and space environments discussed throughout the paper reveals that no single modulation or coding method universally dominates. Instead, optimal choices arise from matching physical-layer techniques to the operating environment, link constraints, and system objectives. The following guidelines summarize the key considerations for FSO system designers:

- Matching modulation order to channel stability: High-order or multi-carrier schemes perform well in stable conditions, while low-complexity and intensity modulated schemes are more effective in environments dominated by turbulence or platform motion.
- Prioritizing pointing-error resilience where mobility is high: Aerial platforms and long-distance space links require modulation formats that remain robust

under misalignment. Schemes with inherent energy concentration, such as PPM variants or SIM, offer better performance under pointing jitter.

- Balancing spectral efficiency and power efficiency based on mission goals: Terrestrial links often prioritize spectral efficiency due to limited bandwidth, whereas space and aerial systems frequently prioritize power efficiency because of strict SWaP constraints or long link distances.
- Adaptive strategies when channel variability is high: Terrestrial and aerial FSO channels benefit from adaptive modulation, variable-rate coding, and channel-aware selection schemes to maintain acceptable BER under rapidly changing weather and pointing conditions.
- Strong FEC where channel impairments dominate: When turbulence, scintillation, or geometric loss are significant, robust coding provides substantial gains, especially for deep-space or high-availability terrestrial links.
- Considering hardware constraints early in the design process: SWaP limitations restrict modulation and coding choices in aerial systems, while space systems face thermal and pointing budget limits. These constraints shape feasible detector choices, aperture sizes, and allowable DSP complexity.

- System-level metrics for final design selection: BER is essential, but overall performance depends on combining multiple indicators: spectral efficiency, energy consumption, implementation complexity, latency tolerance, and long-term reliability.

Integrating these guidelines with the comparative radar charts in Fig. 8, Fig. 9 and Table II enables a unified view of how each modulation and coding technique performs across key metrics. The optimal selection of modulation and coding schemes thus involves balancing three fundamental resources: power, bandwidth, and complexity. C_{sys} denotes a generic system-level quality-of-service (QoS) metric, which may represent achievable rate, spectral efficiency, outage-limited throughput, latency-aware utility, or other application-dependent performance measures, evaluated under the assumed FSO channel conditions and receiver architecture. We can express the optimization problem as:

$$\begin{aligned} & \max_{\text{mod, code}} C_{sys}(\text{mod, code}) \\ \text{subject to: } & P_t \leq P_{\max}, \\ & B_{\text{req}} \leq B_{\max}, \\ & \text{Complexity} \leq C_{\max}, \\ & \text{BER} \leq \text{BER}_{\text{target}}, \end{aligned} \quad (42)$$

where the constraints reflect platform-specific limitations (e.g., eye-safety power limits, available bandwidth, processing capability, and quality-of-service requirements).

VII. SECURITY IMPLICATIONS OF MODULATION AND CODING IN FSO SYSTEMS

Beyond reliability and efficiency, modulation and coding schemes in FSO systems also influence the physical-layer security of the link. Due to the narrow beam divergence and inherent spatial confinement of optical carriers, FSO links already provide a high level of directionality-based security compared to RF systems [208]. However, the choice of modulation format can significantly affect the susceptibility to eavesdropping and intentional jamming [209]. The degree of protection varies with the deployment scenario.

In terrestrial links, security is primarily constrained by scattered-light leakage and the ease of physical access to the propagation path; intensity-based schemes such as OOK and PAM are more susceptible to intercept-and-replay attacks because their waveforms are directly observable from residual reflections, whereas heterodyne detection improves stealth by reducing secrecy outage probability [210]. Aerial FSO links experience additional risks from platform motion, pointing jitter, and active jamming [211]; here, differential phase formats such as DBPSK and DQPSK offer stronger resilience because they avoid explicit carrier phase recovery and complicate coherent interception, while multi-carrier schemes (e.g., SIM and OFDM) can suppress narrowband jamming through subcarrier allocation [212]. Satellite and deep-space links benefit from extremely narrow beam divergence and large propagation distances, making passive interception geometrically infeasible,

although ground-to-space uplinks remain vulnerable to blinding and high-power jamming [213], [214]. Coherent PSK/QAM formats can also be used to enhance security due to the stringent phase and synchronization requirements needed for unauthorized detection [142].

On the coding side, strong FEC mechanisms such as LDPC, Turbo, Polar, and rateless fountain codes not only improve reliability but also enhance secrecy by creating a decoding-threshold gap between legitimate receivers and potential eavesdroppers [206], [215]. Ultimately, the achievable security level depends on link geometry, range, platform stability, beam divergence, and environmental conditions [216]. For instance, a short-range, well-aligned terrestrial FSO link may offer stronger practical security than a long-range LEO downlink, whereas GEO inter-satellite links can be effectively immune to passive interception. These considerations highlight the need to integrate physical-layer security objectives into the joint modulation-coding design process, where interception and jamming risks are more pronounced.

VIII. EMERGING TRENDS AND FUTURE DIRECTIONS

With the continuous advancement of FSO communication technology, driven by the demand for higher data rates, greater reliability, and better adaptability to diverse communication environments, several emerging trends and future directions are gradually taking place. These trends aim to enhance the performance of FSO systems, overcome existing limitations, and ensure their integration with other communication technologies. The following are some of the key emerging trends and future directions in the field of FSO communication, focusing on the development of channel coding and modulation formats in terrestrial and non-terrestrial laser communications.

A. Modulation Formats

FSO communication systems are rapidly evolving beyond traditional OOK towards more advanced modulation strategies, aiming to maximize throughput, enhance resilience, and adapt to diverse atmospheric conditions. Several key trends are shaping the future of FSO modulation.

First, adaptive modulation is gaining traction as a core strategy. By dynamically adjusting the modulation format based on real-time channel conditions, adaptive modulation enables FSO links to maintain high reliability and spectral efficiency under varying turbulence and weather scenarios [15]. Second, polarization modulation and OAM modulation are being increasingly explored. Polarization modulation leverages the relative stability of the polarization state through atmospheric channels, providing greater robustness against turbulence, scattering, and misalignment. This makes it particularly suitable for terrestrial FSO systems and short-range communication [217]. In contrast, OAM modulation exploits orthogonal twisted light modes to multiplex multiple parallel data streams, offering substantial capacity gains. OAM techniques are especially promising for satellite and long-range aerial FSO systems, where atmospheric disturbances are less severe compared to terrestrial environments [218].

TABLE X
MACHINE LEARNING IN FSO MODULATION: TAXONOMY, APPLICATIONS, AND TRADE-OFFS

Ref.	ML Category / Technique	Application & Key Benefit	Limitations & Challenges	Typical Data Requirement
[235]	Deep Neural Network (DNN) for Adaptive Modulation Selection	Adaptive modulation selection for mixed FSO/MIMO-MMW in 5G/6G networks; dynamic adaptation based on atmospheric conditions	Training convergence sensitive to hyperparameters; requires labeled data across diverse turbulence scenarios	Meteorological dataset with C_n^2 measurements (525,000 observations over 11 months)
[236]	Supervised Learning (ANN, KNN, Decision Tree)	Performance prediction and quality assessment for Ro-FSO under fog conditions; ANN achieves best RMSE for SNR prediction	Limited generalization to untrained weather conditions; model complexity increases with modulation order	Historical weather data with corresponding SNR/BER measurements (8 years dataset used)
[237]	Deep Learning-based Blind Detection (Fully Connected DNN and FCNN)	CSI-free detection for OOK modulation in turbulent FSO channels; eliminates pilot overhead and works faster than ML with perfect CSI	Requires retraining for different turbulence regimes; performance sensitive to hyperparameter tuning	Training datasets of 5,000 and 10,000 vectors of 512-bit sequences under weak, moderate, and strong turbulence
[238]	End-to-End Autoencoder with Two-Stage Training	Joint optimization of modulation/demodulation for UAV-to-ground FSO under strong turbulence (Rytov variance up to 3.5)	Training convergence sensitive to hyperparameters; channel model differentiability requirement	Pre-training followed by end-to-end training on 2^{14} (~16,000) randomly generated messages over 2000 iterations

TABLE XI
MACHINE LEARNING IN CHANNEL CODING FOR FSO: TECHNIQUES, TRADE-OFFS, AND TRAINING COMPLEXITY

[246]	Feedforward Neural Network Decoder for (16,8) Polar Codes with tanh-based Modified LLR	Approaches SCL (L=8) decoder BER performance over Gamma-Gamma turbulence; turbulence-stable (trained at one σ_R^2 works across weak/moderate/strong); eliminates CSI requirement	Limited to short code length (N=16); decoder architecture (128-64-32) must be redesigned for different code parameters; tanh scaling parameter requires tuning	Training set: all $2^8 = 256$ possible codewords; 2^{20} epochs (~1M iterations); testing: 10^6 codewords per SNR point
[247]	CNN-based Decoder (Conv+Pooling layers) for Polar Codes in ACO-OFDM FSO	Achieves $\sim 10^2$ BER improvement over conventional SC/BP decoders at 4-QAM under strong turbulence; jointly performs decoding and demodulation without separate equalization	Performance degrades significantly at higher modulation orders (16-QAM, 64-QAM fall between SC and BP); requires retraining for each turbulence regime; high PAPR issue not addressed	Code length N=32, 64; rate=0.5; convolutional architecture with kernel size 3, stride 1; multiple pooling layers; GPU acceleration required
[238]	End-to-End DNN Autoencoder (12,4) with Two-Stage Training (Pre-training + End-to-end)	Outperforms PPM+ML by ~12 dB; reduces decoder time by 51.3% vs ML; achieves HD-FEC threshold (BER $< 3.8 \times 10^{-3}$) at 22 dB SNR under Rytov variance up to 3.5; no CSI needed	Black-box nature limits interpretability; hyperparameter sensitivity (batch size, hidden size); coding efficiency decreases with higher n; requires separate training for each turbulence regime	Training: $2^{14} \approx 16,000$ random messages; 2000 iterations with Adam optimizer; Testing: 2^{18} messages; batch size 128; hidden layers: 32 neurons
[248]	Support Vector Machine (SVM) for LDPC-coded NRCPE Hybrid GMSK-PPM in MIMO/FSO	Kernel-based binary classifier eliminates CSI requirement; achieves near-optimal performance without pilot overhead; diversity gain of 81-90% BER suppression for MIMO (2×2, 2×3) vs SISO	Requires optimal beam width determination through gradient descent; SVM hyperplane computation complexity; limited to binary classification (bit 0 vs bit 1); model retraining needed for channel variations	Two-phase: (1) Training with labeled satellite downlink data to extract support vectors and bias; (2) Validation on unlabeled test set; complexity depends on number of support vectors

Moreover, ML is emerging as a transformative enabler for FSO modulation. ML techniques empower systems to predictably adapt modulation schemes, improving robustness, efficiency, and noise resilience under dynamic channel conditions. In recent studies, ML has been applied in various roles, such as modulation format selection in adaptive MIMO-OFDM and spatial modulation systems [219], [220], modulation and coding scheme (MCS) optimization for dynamic spectrum and SNR conditions [221], [222], and signal demodulation and classification, especially under noisy and distorted environments, using deep learning and dataset-driven training [223], [224]. Machine learning techniques for FSO modulation can be systematically categorized by their learning paradigm and operational role, as summarized in Table X. Supervised learning approaches—including deep neural networks (DNNs), artificial neural networks (ANNs), k-nearest neighbors (KNN), and

decision trees—dominate adaptive modulation selection and performance prediction tasks [225], [226]. These methods leverage historical channel data but require extensive labeled datasets and face generalization issues in untrained conditions. Deep learning-based non-coherent detection techniques eliminate channel estimation overhead, though they suffer from interpretability constraints [227]. End-to-end autoencoders jointly optimize transceivers for strong turbulence, yet their convergence is sensitive to hyperparameters and they require differentiable channel models [228]. Reinforcement learning (RL) techniques enable adaptive selection through environmental interaction but contend with convergence times that may exceed channel coherence and safety risks during online training [229]. A fundamental challenge across all paradigms is poor generalization to diverse link conditions, exacerbated by the difficulty of collecting representative channel data spanning rare atmospheric

TABLE XII

LITERATURE ON MODULATION, CODING, AND TURBULENCE MODELS HIGHLIGHTING PERFORMANCE METRICS AND ENHANCING TECHNIQUES IN FSO SYSTEMS

Ref.	Coding	Modulation	Turbulence Model	Performance Metric	Enhancing Technique
[100]	Block, Convolutional, Turbo	OOK	LN	BER	Diversity
[27]	-	Multi-pulse PPM	GG	SER, Outage probability	Diversity
[62]	Turbo	OOK, 8-PPM	Non fading	BER	-
[116]	Repetition	BPSK-SIM	GG	BER	Diversity
[126]	Random coding	PPM	LN, Exponential, GG	Outage probability	Diversity
[145]	oFEC, Staircase	QPSK, 8-QAM, 16-QAM	Non fading	BER	-
[149]	-	PPM-BPSK-SIM	GG, LN	BER	Diversity
[252]	LDPC	PAM	GG	BER	Diversity
[181]	Convolutional	PAM	GG	BER	Diversity
[182]	Alamouti STBC	PAM	Generalized Gamma	BER	Diversity
[183]	QC-LDPC	16-QAM	GG	BER, OSNR	-
[184]	Convolutional	DPIM, BDPIIM	GG	BER	-
[199]	Convolutional	M-QAM	Hufnagel-Valley	Throughput, Energy efficiency	HARQ
[202]	LDPC	MPAM, MPPM	Non fading	BER	ACM
[206]	BCH, LDPC	OOK	GG	BER	Diversity
[253]	STBC, V-Blast	BPSK, QPSK, 8-PSK	Von Kármán	BER	Diversity, OAM
[254]	RS	PPM	LN, Exponential	BER	-
[255]	-	OOK, PPM, PAM, BPSK, QPSK, DPSK	GG, LN	BER	-
[256]	-	PPM	GG	SER, Channel capacity	Diversity
[257]	LDPC	OOK	GG	BER	Diversity
[258]	Convolutional	PPM	GG	BER	-
[259]	-	PDM-DPSK-MPPM	GG	BER	-
[260]	-	BPPM	GG, LN	Outage probability	Diversity
[261]	Convolutional	MPPM	GG	BER	-
[262]	-	OOK	LN	Outage probability	Diversity
[263]	-	BPSK, DPSK	GG	BER	Diversity
[264]	-	QPSK	Non fading	BER, Outage probability	Diversity
[265]	Alamouti STBC	OOK	Málaga distribution	BER, Outage probability	Diversity
[266]	-	OOK, DPSK, DQPSK	GG	BER	Diversity
[267]	-	PPM	GG	BER	Diversity
[268]	RS	M-QAM-OFDM	GG	ABLER	-
[269]	LDPC	OOK	LN, GG	BER	-
[270]	STBC, O-STBC, STC, STTC	OOK	GG	BER, Outage probability	Diversity
[271]	STC	QPSK	Kolmogorov	BER	Diversity, OAM
[272]	Turbo	16-ary shift keying	Von Kármán	BER	OAM, ML
[273]	-	16 QAM-OFDM	Hill-Andrews	BER	OAM, ML

events. Furthermore, implementing low-latency inference under strict power budgets remains a critical constraint for deployment on resource-limited platforms like small satellites and UAVs.

B. Channel Coding

Emerging trends in channel coding for FSO systems are moving toward specialized schemes that address the unique challenges of terrestrial, aerial, and satellite platforms. These trends emphasize the use of advanced hardware, such as single-photon detectors, to handle complex and dynamic channel conditions. For satellite FSO systems, coding techniques are being developed to resist interference, noise, and signal attenuation, while terrestrial and aerial platforms focus on coding that adapts in real time to atmospheric turbulence and obstructions.

Over the past few years, machine learning has revolutionized channel coding by introducing intelligent, data-driven algorithms that extend and enhance traditional error-correction schemes. Deep learning techniques such as convolutional and recurrent neural networks have demonstrated robust decoding capabilities under harsh and fluctuating channel conditions [230], [231]. Reinforcement learning has also been explored for adaptive bit-flipping decoding strategies, learning

to correct errors based on feedback from channel environments [232]. Furthermore, supervised learning approaches enable the replacement or enhancement of fixed FEC codes with flexible, learnable decoding architectures that generalize across code types [233]. These advances allow for end-to-end learning of encoder-decoder pairs, error-tolerant IoT designs, and even integrated sensing and communication sequence optimization [234], [235]. Neural network decoders, including those for polar codes and end-to-end autoencoders, demonstrate turbulence-robust performance and can achieve significant bit error rate improvements without relying on explicit channel state information. However, these data-driven approaches are constrained by substantial training data requirements, often necessitating millions of codeword examples spanning diverse channel conditions to ensure generalization. Furthermore, their practical deployment on resource-constrained platforms, such as satellites and UAVs, is challenged by high computational latency, significant memory footprints, and strict power budgets, often mandating model compression and radiation-hardened hardware implementations. Consequently, while ML-aided coding can provide notable performance gains, its adoption requires a careful evaluation of the inherent trade-offs between improved robustness and the associated data, training,

and hardware overheads. A comparative analysis of these techniques, highlighting their trade-offs and implementation challenges, is summarized in Table XI

Deploying ML-aided coding and modulation schemes on resource-constrained FSO platforms presents significant implementation challenges that span hardware, data, and operational domains. For satellite terminals, neural network models must comply with severe onboard constraints, including strict power budgets, limited memory, and the need for radiation-tolerant processors with lower computational throughput than commercial counterparts [239]. This requires the use of model compression techniques, such as quantization, pruning, and knowledge distillation, to reduce the memory footprint and enable feasible inference latency [240]. UAV-based FSO systems face distinct challenges from platform vibration, thermal fluctuations affecting laser diodes, and intermittent connectivity that complicates real-time model updates [34]. While terrestrial high-rise links must adapt to slow-varying channel changes induced by building sway, the convergence time for online retraining (thousands of iterations) may exceed tolerable outage windows. Furthermore, collecting diverse and representative training data is exceptionally difficult; satellite downlink datasets differ fundamentally from terrestrial horizontal links in turbulence statistics, and availability of ground truth [241]. These challenges make transfer learning and domain adaptation promising yet underexplored aspects for cross-platform model portability in FSO systems.

FSO systems on all platforms face a trade-off between assuring reliability and high-performance communication and adding complexity to the system. Optimizing for the specific operational and environmental difficulties that each platform faces while limiting the complexity level is essential to the design of effective FSO systems. Furthermore, while equalization is not always necessary, it can be advantageous in situations with low elevation angles [36]. The primary goal is to develop modulation/coding schemes acceptable for signal transmission (with certain QoS and BER criteria) and applicable under different attenuation situations. Table XII presents efforts in FSO literature in modulation and coding schemes for different channel models, the investigated performance metric, and whether an enhancing technique is used or not.

IX. CONCLUSION

This survey has offered a comprehensive examination of modulation and channel coding techniques for FSO communication systems across terrestrial, aerial, and space environments. By systematically comparing system characteristics, channel models, and performance metrics, the survey highlights the unique challenges posed by each deployment scenario and the corresponding requirements placed on physical-layer design. Terrestrial links are dominated by atmospheric turbulence and weather-induced impairments, aerial platforms encounter severe pointing errors and strict SWaP constraints, and space systems demand ultra-precise alignment and high power efficiency over long link distances. These distinctions underscore the necessity of environment-aware selection of modulation formats and error-correction schemes. The design guidelines

provided serve as a concise decision-making framework that can assist system engineers in aligning modulation and coding choices with mission constraints, channel variability, and system-level objectives. Emerging trends indicate a clear shift toward intelligent, flexible, and resilient FSO systems. As future communication networks evolve toward higher data rates, tighter reliability targets, and more diverse deployment environments, the insights consolidated in this survey provide a foundation for designing robust, efficient, and scalable FSO communication systems.

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